Limiting For-Profit Provision in Nursing Home Markets

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Abstract

We examine whether policies that ban for-profit providers and allow for subsequent takeovers by not-for-profits can be effective at addressing quality shortfalls. We study the US nursing home industry, where quality provision has been a concern for many decades. We document a quality-access tradeoff that makes welfare effects theoretically ambiguous: not-for-profit providers choose higher-quality inputs than their for-profit counterparts but serve wealthier residents, leaving for-profit providers to play an important role in providing access. We estimate a structural model of nursing home demand and supply, revealing that while not-for-profit providers have substantial non-pecuniary motives, in equilibrium, not-for-profit providers target residents with stronger preferences for quality due to their cost disadvantage. Accordingly, not-for-profit takeovers of the lowest quality for-profit providers reduce consumer surplus by $2.3 million a year on average. These results underscore the role that for-profit providers play in expanding access to nursing home care.

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1 Introduction

Quality under-provision is a persistently challenging problem in regulated markets such as in health and education. Due to their profit-seeking motives, for-profit (FP) providers are easy targets for blame, leading to calls by some policymakers to limit their activities. For example, policymakers in California sought to limit the operations of FP colleges (Mello, 2019), while policymakers in New York sought to ban the entry of new and expansion of existing FP nursing homes.[1] Implicit in these prescriptions is the belief that these services are better delivered by not-for-profit (NFP) providers instead, whose objectives extend beyond profit maximisation.

We ask whether these beliefs are justified: should policymakers limit FP provision in certain markets? Adjusting the extent for FP versus NFP provision through changes in certification requirements is a tool in the policymaker's toolbox that should be studied given the limited success of other types of reforms.[2]

Some merit in these ideas is warranted. If NFP providers have non-pecuniary motives which are more aligned with a social planner's, increased NFP provision could lead to increased quality. On the other hand, if FP providers are more cost-efficient, then they could play an important role in expanding access to services. In particular, lower costs allow them to serve at lower prices which facilitates access by poorer people to important services. Lower costs also enable FP providers to operate in markets where revenues are lower due to relatively low regulated prices. Thus, the effects of policies which adjust the extent of FP versus NFP provision are ambiguous and depend on the relative differences between non-pecuniary motives and costs.

We study this question in the context of the US nursing homes industry, an industry plagued by longstanding quality underprovision.[3] Focusing on California, we estimate a structural model of nursing home demand and supply to simulate the effects of increasing NFP provision by having the lowest quality FP nursing homes be taken over by NFP providers. We find that this policy would

[1] These efforts had significant support with associated bills passing the New York state assembly twice (New York State Legislature, 2021a). Additionally, regulations which limited profits to 5% of expenses were legislated in 2021 (New York State Legislature, 2021b).

[2] For example, two broad types of policies aimed at improving quality include pay-for-performance (Cohodes et al., 2023; Fryer, 2013; Gupta, 2021; Alexander, 2020; Finkelstein et al., 2018) and improved provision of information – often through 'report cards' (Dranove, 2011; Dranove and Jin, 2010). Overall results on these policies in both health and education settings are mixed. The key issue is that firms can game payment schemes and report cards to avoid improving quality.

[3] A study report by the National Academies of Sciences, Engineering, and Medicine (2022) provides a summary of the evolution of policy and reforms to improve nursing home quality going back several decades. In their introduction, they reflect on their predecessor report from 1986, which raised serious concerns regarding quality provision. They also highlight how COVID-19 laid bare the continued quality failings in nursing homes. Approximately 40% of all US COVID-19 deaths were among nursing home residents (SteelFisher et al., 2021).

be unsuccessful. On average consumer surplus is reduced by $2.3 million a year. Although NFP providers have nonpecuniary motives for serving more bed-days, their costs for serving Medicaid residents – who make up the largest share of bed-days – are significantly higher than FP providers leading to higher prices and lower quality after takeover. More broadly, these results reflect that FP providers play an important role in expanding access to nursing home care. Their lower costs allow them to effectively serve Medicaid residents, who we find to have weaker preferences for quality.

Our framework and results are motivated by observed patterns of quality and access across FP versus NFP providers in our nationwide sample of nursing homes. For quality, we focus on a nursing home's licensed nursing intensity, while for access we look to the extent which nursing homes serve private payers versus Medicaid payers. Private payers typically pay the private price set by the nursing home out of pocket, while Medicaid residents – whose stays are funded by government – are especially poor as eligibility requires exhaustion of wealth and assets.[4]

Using cross-sectional variation, we find that NFP providers choose higher quality than their FP counterparts. On the other hand, FP providers are more prevalent in poorer markets and serve more Medicaid residents than NFP providers. We also find that markets with lower income have more Medicaid residents, while markets with higher income have more private-payers. Thus, a potential explanation is that NFP providers may be more prevalent in higher income markets because they prefer serving private payers.

In addition to counterfactual simulations, our structural model of nursing home demand and supply allows us to better understand these observed quality and access patterns. On the supply side, nursing homes compete by choosing quality and private prices to maximise an objective that includes profits and nonpecuniary motives for providing care. We allow cost functions and the quantity-of-care motive to vary systematically across NFP and FP providers so that we can capture the extent to which differences in these drive pricing and quality choices. Cost functions also vary systematically with the payer-type quantities served, reflecting that the cost of serving patients of different payer-types varies due to their differing care needs. This allows us to capture payer-type margins that vary at the firm level. These are especially important for explaining the patterns of access and provision. For example, estimates revealing large private-payer margins for NFP providers would be consistent with their prevalence in higher income markets.

An empirical challenge is to separately identify the nonpecuniary motive versus cost function parameters. Larger marginal costs can arise from either larger cost function parameters or larger nonpecuniary motives which incentivize the provider to operate at a higher level of quantity. We distinguish between these by matching variable costs implied from the model with those from

[4]We introduce payer types in greater detail, including Medicare payers, when we discuss the setting in Section 2.

firm-level cost report data. Fixing quantities, any differences must be rationalised by differences in cost function parameters.

Our demand side captures preferences over price, quality, and NFP status and allows them to vary significantly with our rich individual demographics. This allows for demand-driven market segmentation as another source of difference in quality and pricing choices across FP and NFP providers. A rich demand side is also important to allow for flexibility across markets in how NFP providers could meet their quantity-of-care motive. Whether they lower prices or raise quality depends on the relative elasticities in the market.

Our model estimates reveal that NFP providers have a nonpecuniary value for providing an additional bed–day of care of approximately \$33 – 17% of their private price or 21% of their average margin. The corresponding value for FP providers is not significantly different from zero. Regarding cost functions, we find that NFP providers incur around \$75 per bed-day more for serving Medicaid payers compared to FP providers. We attribute these cost differences to a variety of sources including differences in resident characteristics, differences in dimensions of quality not captured by licensed nursing intensity and differences in the operational efficiency of operating nursing homes.

We then use the model to explain the in-sample patterns of quality, access and provision. First, regarding patterns about access, we find that FP providers make significantly larger margins on Medicaid payers compared to NFP providers. These differences in Medicaid margins help explain why FP providers are more prevalent in markets with lower socioeconomic status, where Medicaid residents account for a larger share of demand. NFP providers cannot compete for Medicaid payers as well, due to their higher costs. Instead, they tend to serve markets with higher socioeconomic status, fewer Medicaid payers and more private-payers which are relatively profitable.

Turning to quality, we find that NFP providers choose higher quality both because of their nonpecuniary motive and because their margins on private-payers – who have stronger preferences for quality – are relatively larger compared to their margins on Medicaid payers – who have weaker preferences for quality. Our counterfactual simulations also reveal that, at the observed equilibrium, quality is more responsive to changes in nonpecuniary motives than to comparable changes in Medicaid costs. This reflects the payer composition of the markets in which NFP providers operate. NFP providers are more prevalent in markets with more private payers and fewer Medicaid payers, so changes in Medicaid costs have a smaller effect on their quality choices. In other words, the non-pecuniary motive and standard profit-maximising motive more closely align due to the payer mix which NFP providers face in their markets. These results demonstrate that simply studying differences across FP and NFP providers without understanding broader market demand and supply could misattribute all of the differences to nonpecuniary motives as opposed to the outcomes of standard profit-maximising competition.

Finally, we turn to our policy counterfactuals where we ban the lowest quality FP nursing home and have them be taken over by a local NFP nursing home. Since government providers are included in our pool of NFP providers, this could also be thought of as a government takeover.[5] We find that these takeovers reduce consumer surplus by $2.3 million on average per year. This arises primarily because the NFP provider which takes over the FP provider has to raise private prices and reduce quality to reduce costs. Unlike NFP providers, FP providers serve relatively more Medicaid residents, making NFP providers' higher Medicaid costs particularly consequential. These results suggest that reducing FP provision and increasing NFP provision are unlikely to be successful policies. They also underscore the interpretation that FP providers expand access to nursing home care by providing the market with a product that is better suited to serving residents in markets with lower socioeconomic status.

Related literature

While we combine and draw on tools from several papers and bodies of literature for our model, we view our contribution as primarily empirical. Our paper contributes to two bodies of literature. First, this project adds to the extensive literature on NFP providers by estimating a structural model of NFP and FP providers. One focus of the literature has been on understanding the differences between FP and NFP providers in various regulated markets, such as the hospital industry (Newhouse, 1970; Norton and Staiger, 1994; Sloan, 2000; Ballou and Weisbrod, 2003; Horwitz and Nichols, 2009; Tom and Jacobson, 2010; Bayandir, 2012; Dranove et al., 2017; Garthwaite et al., 2018), nursing home industry (Chou, 2002; Grabowski and Hirth, 2003; Ballou, 2005, 2008; Grabowski et al., 2013; Lu and Lu, 2021), and postsecondary education industry (Deming et al., 2012). However, these papers usually approach these differences from a reduced-form perspective, focusing on identifying differences in observable outcomes. For the nursing home industry, earlier works typically find that NFP providers exhibit some form of altruistic motive, typically through better-quality care in some form. However, as Weisbrod (1988) discusses, in general, NFP providers are not well understood.

Reduced-form findings of better-quality care alone are not informative about what is different about NFP providers, especially in an equilibrium with competing FP providers. A structural model allows us to consider a broad range of factors in a coherent way. Our findings reveal that while not-for-profits do have nonprofit motives, they provide higher quality also because of a market segmentation strategy.

In addition, we also document novel patterns of access and market segmentation and provide a complete explanation through the lens of the spatial distribution of payer-types and suitability

[5] Around 20% of NFP providers in our pool are government providers.

of firm types for particular payer-types. We expect our results to generalize to some extent; similar patterns of quality and access are seen in the postsecondary education industry (Deming et al., 2012).

Second, this paper contributes to the literature on long-term care in the US (Cheng, 2023; Ching et al., 2015; Einav et al., 2022; Gandhi, 2023; Gandhi and Olenski, 2024; Gandhi et al., 2025; Harrington et al., 2017; Lin, 2015; Olenski, 2022). A number of papers study determinants of quality and find that Medicaid reimbursement rates (Nyman, 1985; Grabowski, 2001; Hackmann, 2019) and nurse staffing ratios (Harrington et al., 2000) are important. We build on Hackmann's (2019) framework, enriching the demand and supply side of the model to specifically address differences across FP and NFP providers. Other sets of papers find that FP ownership (Chou, 2002; Grabowski and Hirth, 2003) and private equity ownership (Gupta et al., 2021) result in lower quality. These results suggest that if FP providers' inferior quality comes from investing less in quality determinants, limiting FP provision and increasing NFP provision may be welfare-enhancing. However, if FP facilities provide access to care to lower socioeconomic status populations who would not have received care otherwise, reducing FP provision may leave a void in these communities. Li et al. (2015) document that racial and ethnic minorities are more likely to obtain care from facilities with FP ownership. Our project is the first to document and explain the quality access trade-off of NFP and FP provision in a unified framework.

2 Setting

2.1 Not-for-Profit Organisations

NFP organisations are organized and operated exclusively for specific purposes and must not inure any of the earnings to any private shareholder or individual (Internal Revenue Service, 2023).[6] Under section 501(c)(3) of the Internal Revenue Code, qualifying organisations are exempt from federal income tax and can receive tax-deductible donations. They may also qualify for state and local tax exemptions.[7]

NFP organisations must file annual reports (Form 990 or its variant) which include information about revenue, expenditure, and income. Also, NFP organisations may be subject to audits by the IRS or have to submit an audited financial statement depending on specific state laws.[8] We take the effectiveness of this regulation as given and treat NFP and FP providers as distinct firm types.

[6]Specific purposes include charitable, religious, educational, scientific, literary, testing for public safety, fostering national or international amateur sports competition, and preventing cruelty to children or animals.

[7]See https://www.irs.gov/charities-non-profits/exempt-organizations-what-are-employment-taxes

[8]See https://www.councilofnonprofits.org/running-nonprofit/nonprofit-audit-guidec/state-law-nonprofit-audit-requirements

2.2 Nursing Homes

We examine Medicare-certified Skilled Nursing Facilities to study the US nursing homes industry. Skilled nursing facilities primarily provide skilled nursing care and related services for residents who require medical or nursing care and are not primarily for the care and treatment of mental diseases or personal non-nursing care (42 U.S. Code § 1395i–3). Specifically, nursing homes provide two services. First, they provide medical care to individuals who require short-term rehabilitative stays after hospitalization. Second, they provide long-term care services to individuals who require daily nursing care and additional therapy care. This can include young people with significant disabilities.

Table 1 reports summary statistics of nation-wide nursing homes. We observe that 30% of Medicare-certified nursing homes are NFP, while this share has decreased over time. On average, FP facilities are slightly larger.[9] It is generally understood that for-profit nursing homes tend to operate as part of larger chains (Harrington et al., 2012), but the exact extent of this has been difficult to quantify because ownership of for-profit nursing homes is notoriously opaque.[10] Finally, there is little entry and exit, with more in the for-profit sector. This may be explained by NFP providers having limited financing options: they are legally banned from accruing earnings to any private shareholder or individuals, and thus limited from equity financing. These differences in ownership along with differences in payer-types served are suggestive of the different business models adopted by FP versus NFP nursing homes.

Table 1: Summary Statistics - Nationwide Providers

	NFP	FP
Number of Providers in 2000	5,889 (34.7%)	11,075 (65.3%)
Number of Providers in 2018	4,463 (30.0%)	10,437 (70.0%)
Entries per year	34.1	60.6
Exits per year	86.5	114.0
Bed size	98.3	109.1
Average Prop. Medicare in 2018	13.4	13.2
Average Prop. Medicaid in 2018	51.2	63.9

Note: Data in this table are sourced from our facility-level data covering 2000-2018 and includes all states.

[9] Regarding facility ownership, a diverse set of organisations own NFP nursing homes. Of the 4,463 NFP providers in 2018, 953 providers are government-owned, 502 are operated by religious organisations and the remainder by private, NFP organisations.

[10] To improve transparency the Biden Administration recently introduced reforms to increase reporting and monitoring of nursing home ownership (Centers for Medicare and Medicaid Services, 2023).

Reimbursement

Nursing homes are primarily reimbursed on a prospective price-per-day basis. Residents can be distinguished by the funding source for their stay, or their "payer-type". In this paper, we focus on three payer-types: Medicare, private, and Medicaid.[11] Medicare covers post-acute care after hospitalisation for up to 100 days. The first 20 days are fully covered, while there is a copayment for 21-100 days.[12] After residents exhaust those 100 days, they pay the private price set by the nursing home, unless they qualify for Medicaid. As long-term care insurance take-up is relatively low (Brown and Finkelstein, 2007), much of the payment comes from out-of-pocket. Finally, if residents exhaust their financial resources and become unable to pay prices out-of-pocket, Medicaid covers their nursing home expenses.

We report observed prices in Table 2 in California, as MDS and federal data do not report on private prices. Medicare rates are set by the Centers for Medicare & Medicaid Services and take into account factors such as geographic adjustments and expected facility population care needs. Because Medicare patients are generally sicker, Medicare tends to pay the most (table 2). Medicaid prices are set at the state level and are relatively low and there are no explicit adjustments for expected facility population care needs. Given the prospective nature of Medicaid and Medicare prices, we think of nursing homes as taking them as given.

Table 2: Nursing Homes in California in 2010

Panel A: Quantity	Provider Count	Average Bed-Days	
FP	535	36,845	
NFP	74	27,799	
Panel B: Payer Distribution	Medicare	Private	Medicaid
FP	0.189	0.105	0.568
NFP	0.171	0.300	0.336
Panel C: Average Price per Bed-Day	Medicare	Private	Medicaid
FP	227.2	217.9	203.0
NFP	251.7	243.7	196.0

Note: We use California data here as data about private-pay residents is only available from state cost report data and our structural estimation will be based on California from the years 2000 to 2010.

[11] These are not exhaustive, but make up 87% of stays in California. The remainder is from other government sources such as Veteran's Affairs

[12] Since the copayment is facility invariant, we ignore the role of the copayment in our analysis.

On average, in both our nationwide and California data, the largest share of bed-days across both FP and NFP providers are from Medicaid payers (Tables 1 and 2). This arises because Medicaid covers the majority of long stays. Medicare only covers 100 days and nursing home care is expensive, meaning that without long-term care insurance, residents cannot afford to pay out-of-pocket for long. Notably, FP providers tend to serve larger shares of Medicaid residents than their NFP counterparts. This suggests that FP providers play an important role in providing access to care for poorer residents. We explore these access patterns in greater detail when we turn to our motivating evidence.

Residents

Residents vary significantly in their characteristics – including care needs and distance from prior residence – which we allow to govern preferences in our model. Variation in resident characteristics in part arises from the two distinct functions of nursing homes – short-term rehabilitation and long-term care. For example, length of stay varies dramatically with the median stay being 30 days, the average stay being 148 days and the 90th percentile being 410 days in California. The variation in resident characteristics is also reflected in systematic differences in population characteristics across FP versus NFP providers in California and across payer types.

Table 3 demonstrates payer-type specific individual characteristics by provider type.[13] Residents at NFP providers are typically older and stay longer. A resident's activities of daily living (ADL), clinically complex scale (CLN), and cognitive performance scale (CPS) measures non-medical needs, medical care needs, and cognitive performance scale, with higher scores representing higher need of assistance.[14] Residents of FP providers have higher ADL, CLN, and CPS - higher measured care needs. Given systematic differences in the populations choosing FP nursing home care and NFP nursing home care, and to flexibly model substitution patterns we incorporate these individual characteristics for estimating demand.

[13] We provide t-tests for within provider-type and across payer-type comparisons in Table A2 in Appendix A.

[14] ADL ranges from 4 to 18, CLN ranges from 0 to 3, and CPS ranges from 0 to 6

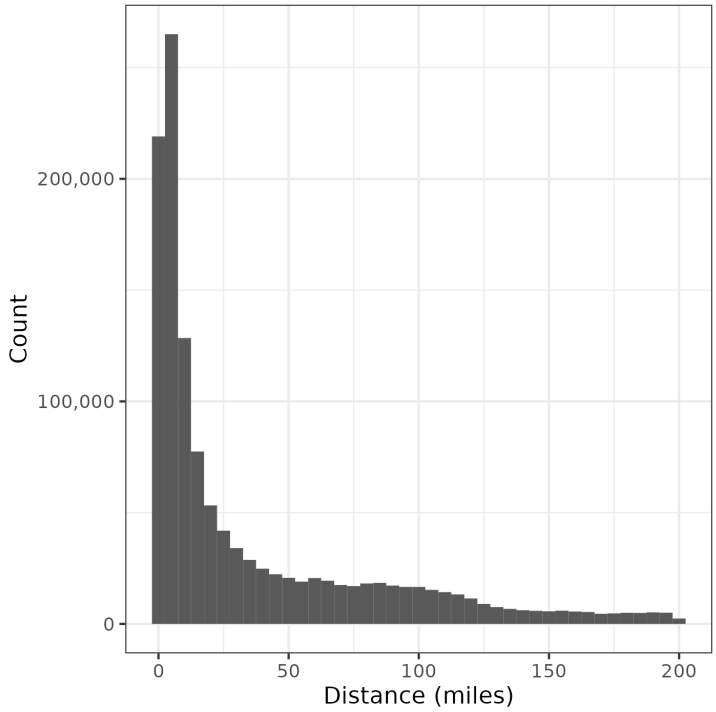


Figure 1: Histogram of distance between a resident's prior residence and nursing home

Table 3: Payer-type average individual characteristics by provider type

Provider Type	Payer Type	Age	Length of Stay	ADL	CLN	CPS	Has Alzheimer's
FP	Medicaid	81.7	466.4	12.32	1.18	2.62	0.091
FP	Private	81.9	92.2	13.01	1.39	1.86	0.047
FP	Medicare	81.9	35.9	12.86	1.38	1.55	0.041
NFP	Medicaid	84.6	582.1	11.84	1.04	2.54	0.096
NFP	Private	84.3	157.0	11.87	1.19	1.71	0.052
NFP	Medicare	83.5	30.4	11.63	1.13	1.30	0.035
Std. Dev		7.89	296.76	3.43	1.00	1.83	0.22

Note: A resident's 'payer-type' for this table is determined by the payer-type that made up the majority of their stay.

Figure 1 plots a histogram of the distance between a Californian resident's prior residence and their nursing home using our individual level data from 2000-2010. Much of the mass is concentrated towards the left, with approximately 10% of stays being in the same zip and the median distance being 12.8 miles. This suggests that nursing home residents have a preference for being close to their prior residence which we allow for in our demand model. On the other hand, there is also a long tail leading to the average distance being 36.3 miles.

2.3 Continuing Care and NFP Business Models

One striking difference between FP and NFP providers in California is the proportion of private payer days served with NFP providers serving almost 3 times as much as FP providers (table 2). This may be explained by the prevalence of continuing-care arrangements among NFP providers. Under these arrangements, providers operate upstream independent or assisted living facilities alongside skilled nursing facilities, allowing residents to transition to more intensive care as their needs evolve.

Since independent and assisted living facilities provide significantly less medical care, Medicaid and Medicare play a much smaller role in funding care at these facilities. By the same token, residents in these arrangements are likely to cover more of their skilled nursing stays as a private-payer.

The value of the skilled nursing facility in continuing care arrangements is ensuring there is a place in the community where they can receive long-term care with more medical care. The importance of the continuing care arrangement was confirmed by the CEO of a NFP nursing home who told us that NFP providers are unlikely to acquire or construct new nursing home facilities, unless they can also acquire or construct an independent living or assisted living facility.

Since independent and assisted living facilities are primarily privately funded, the sector is much less regulated, with little comprehensive data. Zarem (2010) cites that 82% of such providers are NFP and 18% are FP. More recently, Nelson (2018) cites that 78% of such providers are NFP and 22% are FP.

These arrangements do not mean that the skilled nursing facilities exclusively serve residents of the associated upstream facilities. We hand-collected data on whether each nursing home that we could look up from our California nursing home data was likely to be associated with an independent living or assisted living facility.[15] We found that the average and third quartile share of Medicaid payers at nursing homes associated with such facilities is 39% and 66% respectively.

These arrangements may help explain differences in payer composition, costs, and unobserved dimensions of quality across FP and NFP providers. Although we cannot explicitly model continuing-care arrangements due to data limitations, we draw on this institutional context when interpreting our structural estimates.

[15] Our hand-coding relied on nursing homes in our data, but based on recent information, rather than between 2000-2010. Our coding aligns with the cited earlier public reports. We observe that 89% of NFP nursing homes and 19% of FP nursing homes that we were able to find information for had upstream facilities.

3 Data

We use three sets of data. Most of the descriptive analysis uses data covering all US states between 2010 and 2018. Structural estimation will focus on resident stays in California from 2000 to 2010.

Resident Data The basis of our resident data is the nursing home Minimum Data Set (MDS). As per federal law (42 CFR §483.20), all nursing homes that are eligible to provide care and be reimbursed by CMS either through Medicare or Medicaid are required to complete assessments about their residents. Assessments are essentially an extensive survey that results in a dataset that provides a comprehensive 'assessment' of the resident's health, mental condition and cognitive abilities. Assessments are conducted upon admission, discharge and throughout the resident's stay at least quarterly. This data allows us to construct a panel of resident stays which is the basis of our structural estimation. It also provides zip-codes of where residents previously lived which allows us to construct zip-centroid distances.

Although the MDS also contains information about resident payer-types, it has been noted in the nursing homes literature that the payer-type fields in this data have inaccuracies (Grabowski et al., 2008; Hackmann, 2019; Cheng, 2023).[16] Following the literature, we use Medicare claims data from the Medicare Provider and Analysis Review (MedPAR) files to identify when residents are covered by Medicare. To identify when residents are covered by Medicaid, we draw upon the Medicaid Analytic Extract (MAX) files. For 2001-2005 we have access to the MAX Long Term Care file and for 2006-2013, the MAX Personal Summary file. We also supplement this with data from the Medicare Beneficiary Summary file which we have from 2000-2019.

Facility Data We obtain facility characteristics and location data from LTCFocus who derive their data from the Centers for Medicare and Medicaid Services (CMS) Online Survey, Certification and Reporting database.[17] Since data on private prices are not collected by any regulators we follow the literature and infer private prices by dividing total private-pay revenues with total private-payer quantity from California nursing homes cost report data (Huang and Hirth, 2016; Gandhi, 2023). We also source facility financial data (costs) from California cost report data.

[16] There does not appear to be a comprehensive understanding about the source of inaccuracies, but there appear to be two types of issues. As described in Cheng (2023) (section B.2) one issue may arise from nursing home uncertainty about the payer-type. A second issue may arise from payer-types not being updated in the data from the admission assessment, even when the payer-type may have changed in a subsequent assessment (Grabowski et al., 2008).

[17] LTCFocus is a Gerontology and Healthcare Research center at Brown University. LTCFocus Public Use Data is sponsored by the National Institute on Aging (P01 AG027296) through a cooperative agreement with the Brown University School of Public Health.

Geographic and Demographics Data We source our demographic and geographic data from various datasets from the census bureau. It is worth noting that demographic data, including populations by age at the zipcode level only exist in the 2000 and 2010 decennial census and from the American Community Survey from 2010 onwards.[18] These data exist at the county level for our whole sample period.

4 Motivating Evidence: The Quality-Access Trade-off

4.1 Quality Provision

Following previous literature, we measure nursing home quality by the sum of registered and licensed practical nursing hours per resident day - 'total' nursing hours per resident day (Hackmann, 2019; Lin, 2015, 2014; Friedrich and Hackmann, 2021). Registered nurses are typically regarded as higher skilled nurses with more qualifications compared to their licensed practical nurse counterparts. Throughout this paper we focus on a nursing home's choice of nursing hours rather than quality outcomes. There are two reasons for this. First, comprehensively and accurately measuring quality outputs is difficult.[19] Second, the role of costs will be important in the structural model and counterfactuals. There is a clear relationship between inputs and costs, less so for outputs and costs. Still, Grabowski et al. (2013) find that NFP nursing homes provide better-quality outcomes than their FP counterparts.

Table 4 shows that FP providers are associated with lower nursing hours per resident day for both nursing types compared to NFP providers, on average. This also holds after controlling for facility, facility population controls and year and zip fixed effects. The association for registered nursing hours per resident day is especially large with FP providers using around 40% (0.2/0.492) less compared to NFP providers.

In appendix A.1, we use the same specification to explore the private prices set by nursing homes and find that NFP nursing homes in California also set higher private prices.[20]

These results are consistent with our interpretation that after accounting for differences in care needs, NFP providers offer a higher quality, higher priced product, while FP providers offer a lower quality, lower priced product. To disentangle whether these choices are driven by preferences for

[18] Strictly speaking, these are termed as zip-code tabulation areas by the U.S. Census Bureau, since data are not directly collected at the zipcode level.

[19] Recent literature has made progress on this issue (Cheng, 2023; Einav et al., 2022); however, quality measurement in these works are ultimately based on short-term mortality relating to a nursing home's role as a rehabilitation center rather than as a long-term care provider.

[20] Private prices are sourced from California cost reports, thus the sample used is different to the nationwide sample used here.

quality, differing costs and/or differing nonpecuniary motives we will turn to a structural model of nursing home demand and supply.

Table 4: Differences in Nursing Inputs

Hours/ resident-day	Registered Nurse	Licensed Practical Nurse
FP (Diff. in Means)	-0.322	-0.083
FP (Regression)	-0.200	-0.055
	(0.009)	(0.009)
Mean of dep. var	0.492	0.853

Note: These results use yearly facility-level data across 2010-2018 and all states. Controls include facility size (in beds), aggregate activities of daily living score, which measures how dependent the population is on carer support for daily needs, aggregate case mix index, which measures how much medical care the population requires and proportion of Medicaid and Medicare residents in the facility. Year and zip fixed effects are included with standard errors clustered at the facility-level.

4.2 Access to Providers and Patterns of Provision

There are two dimensions of access that nursing homes can provide. First, supplying capacity; second, access to certain demographics. As shown previously, FP providers serve a large proportion of the market (Tables 2 and 1).

Second, we find that FP providers systematically serve different markets and different residents compared to their NFP counterparts. In particular, we find that FP providers serve less wealthy residents compared to NFP providers, both across and within markets. In row 1 of table 5 we regress the absolute difference in number of NFP providers to FP providers in a given zip-code on zip characteristics, county and year fixed effects and observe that across markets, NFP providers are more prevalent in higher income zip-codes.

We next turn to comparisons of facilities within markets and find similar patterns. Regressing the proportion of payer types on FP status, facility, facility population controls and year and zip fixed effects we find that FP providers serve larger proportions of Medicaid payers and lower proportions of Medicare and private-payers compared to their NFP counterparts (table 6). To the extent that Medicaid residents cannot afford to pay out-of-pocket, we interpret these results to suggest that even when NFP providers are located in markets with lower socioeconomic status, they

still tend to serve more well-off residents.

Table 5: Location patterns

		log(Median Income)
#NFP - #FP		0.167
		(0.016)
Prop Private Payers Served By	NFP	0.108
		(0.013)
	FP	0.08
		(0.007)
Prop Medicaid Payers Served By	NFP	-0.121
		(0.018)
	FP	-0.141
		(0.011)

Note: Each row is a regression. These results use facility-level data across 2010-2018 and all states. All specifications include year and county fixed effects and control for zip characteristics including logged 15-64 population and logged 65+ population, share of residents who are black, log of the zip area and proportion of the zip which is urban. Standard errors clustered at the county level.

Table 6: Proportion of payer-types served

	Prop. Medicaid Payers	Prop. Medicare Payers	Prop. Private Payers
FP	0.140	-0.060	-0.080
	(0.004)	(0.003)	(0.003)
Mean dep. var	0.60	0.15	0.25

Note: These results use facility-level data across 2010-2018 and all states. Controls include facility size (in beds), aggregate activities of daily living score, which measures how dependent the population is on carer support for daily needs, aggregate case mix index, which measures how much medical care the population requires. All specifications include year and zip fixed effects. Standard errors clustered at the zip level.

Our interpretation of these results is that NFP providers serve a product better suited to private payers, consistent with continuing care (section 2.3), while FP providers serve a product better suited to Medicaid payers. To build support for this interpretation we turn to rows 2-5 of table 5 and study the proportion of private payer and Medicaid residents served by all NFP and FP providers

in a zip-code. FP and NFP nursing homes serve more private-pay residents in markets with higher socioeconomic status and more Medicaid residents in markets with lower socioeconomic status. A 10% increase in median income is associated with approximately 1.1 percentage points more private payers and a decrease in Medicaid payers of a similar magnitude. These facts suggest that the spatial distribution of payer-types is correlated with socioeconomic characteristics. [21]

A structural model is required to close the story. First, from the supply side, it remains to be shown that FP providers prefer serving Medicaid residents and that NFP providers prefer serving private-payers.[22] In other words, that Medicaid residents are more profitable for FP providers and private-payers are more profitable for NFP providers. This requires estimation of a structural model of nursing home supply, since marginal costs are not directly observed. Second from the demand side, evidence for differential preferences for quality and private prices are also important for the interpretation that the NFP provider product is more suited to private payers and that the FP provider product is more suited to Medicaid payers. This will come from the structural model of nursing home demand.

It is worth noting that these spatial patterns could suggest modelling provider location choices as being important. However, we abstract from this for two reasons. First, as table 6 showed, these patterns arise within markets, thus after accounting for resident preferences for distance, a model of price and quality competition that allows for systematic differences by private payers, Medicare and Medicaid payers is sufficient to explain those patterns. Second, extensive margin activity in the form of entry is very limited in California. We observe only 45 facility entries over 2000-2010. Thus, at least for our sample and for our counterfactuals that study government-mandated takeovers of FP providers by NFP providers, location choices are not relevant.

[21] Another potential reason for the observed NFP location patterns is that NFP providers may find it easier to secure donations and funding in wealthier locations. In interviews with industry members we heard that the role of donations and funding is relatively limited. In appendix A.2, we verify this claim and show that most NFP nursing homes do not receive any donations, including those in higher income markets.

[22] In theory, nursing homes could selectively admit residents to serve more of their preferred payer-type. However this is limited by regulation. Gandhi (2023) studies selective admissions by nursing homes. Our interpretation of his work is that the regulations are effective. He does find evidence that nursing homes undertake selective admissions and in particular discriminate against Medicaid-eligible residents. However, the magnitude of selective admissions which Gandhi finds is quite small. Moreover, selective admissions do not affect the arrival rate of preferred payer-types. For these reasons, we regard location patterns as more important and do not deal with selective admissions in this paper.

5 Structural Model

To explain the patterns of quality-access provision and examine the effect of counterfactual policies, we develop and estimate a structural model.[23] The model is a static oligopoly game where providers compete by choosing their private price and quality (total nursing hours per resident-day).[24] Providers draw their cost and demand shocks which are all observed by providers. In the second stage, providers choose private price and quality. Finally, residents choose the nursing home which maximises their utility. The equilibrium concept is Nash equilibrium.

We highlight a few notable features of the model. First, providers have standard motivations from profit, but they may also have nonpecuniary motives. Specifically, we allow for providers to explicitly place value on the quantity-of-care they serve, and this will be a parameter to be estimated. If residents value quality, then a nonpecuniary motive for quantity-of-care served could lead to the provider choosing higher quality. This necessitates the estimation of demand.

Our model allows for rich heterogeneity in both demand and supply. Demand is rich allowing for substantive heterogeneity in preferences for private prices and quality. Heterogeneity in cost is also rich, as different payer-types may have different costs (and margins) and these costs may also differ by firm-type.

Together, these components generate strategic incentives for providers to segment the market and target specific payer types through their price and quality decisions. Whether a firm would prefer to choose a higher quality depends not only on its objectives, but also on whether higher quality attracts more desirable, higher-margin residents. Modelling payer-type margins allows us to explain both quality and access, as firms in markets with more of the desirable payer-types will be relatively profitable.

5.1 Demand

We adopt a similar model of demand as Hackmann (2019). Prior to a potential stay, resident i chooses the nursing home j that maximises their daily utility during the stay.[25] Markets are

[23] To the best of our knowledge, there are no satisfactory policy changes with exogenous variation in the number of NFP versus FP providers which would allow us to explore such counterfactuals.

[24] We model providers who choose their quality by choosing their total nursing hours per resident-day. This follows previous structural work such as Hackmann (2019) and Lin (2015) and a broader literature on the effects of nursing inputs on quality outcomes (Lin, 2014; Friedrich and Hackmann, 2021). We also follow the literature and restrict nursing hours to focus exclusively on higher-skilled nurses - registered nurses and licensed practical nurses but not certified nursing assistants.

[25] Residents implicitly choose the nursing home which maximises the utility of their entire stay which is the product of their daily utility and length of stay. Implicit in this, we assume that length of stay is exogenous. On the demand side this means that residents do not evaluate how their length of stay might change if they chose a different nursing home.

metropolitan statistical area-year pairs in California from 2000-2010. Individual nursing home stays need not adhere to a calendar year and 22% of stays in our data overlap with at least 1 year. We assume that the nursing home commits to the same price and quality throughout a resident's stay. That is, in subsequent years, the nursing home's pricing and quality choices apply only to residents entering in that year. Since residents only decide at the start of their stay, allowing for prices and quality to change for the resident in subsequent years would result in having to treat the resident as fully 'locked in' for subsequent years.[26] Residents who are only Medicare or Medicaid payers over the course of their stay never pay the private price, while those who are private-payers at some stage do. Accordingly, we further distinguish resident utility by private-price exposure - $y \in \{\text{non-priv}, \text{pays priv}\}$.[27] Then an individual's utility is:

$$u_{ijt}^{y} = \beta_{1i}^{d} D_{ij} + \beta_{2i}^{d} D_{ij}^{2} + \beta^{I} \mathbf{1}(j = j(t-1)) + \beta_{i}^{TN} TN_{jt} + \beta_{i}^{p,y} p_{jt}^{priv} + \beta_{i}^{NFP} NFP_{j} + \lambda_{j} + \lambda_{t} + \xi_{jt}^{y} + \varepsilon_{ijt}$$

Coefficients which vary by individual are of the form:

$$\beta_{i}^{k} = \beta^{k} + \sum_{r} z_{ir}^{k} \beta_{r}^{k}$$

with $\beta_{i}^{p,\text{non-priv}} = 0$.

Residents have preferences for distance D_{ij} which are allowed to be nonlinear as figure 1 showed that there is a long tail of residents whose nursing home is far from their prior residence. We allow preferences for distance to vary by age of the resident on entry, realised length of stay and whether the resident has a legal guardian who is not themselves.[28] We allow for inertia in demand through $\mathbf{1}(j = j(t-1))$, reflecting high switching costs once a resident has moved into a nursing home. Since our data span a long period (2000-2010), many residents undergo multiple stays. Inertia could introduce dynamic considerations; however, we assume that residents are myopic so

On the supply side, this means providers' choices cannot affect the length of stay through their quality choices.

[26] Since private price setting is largely unregulated, and nursing homes do not typically advertise their private price rates, to the best of our knowledge it is not clear what happens in practice.

[27] Although Private-pay, Medicare and Medicaid make up the vast majority of residents, some residents are of a different payer-type altogether. Two examples of such payer-types are those who access nursing home care through Veteran Affairs or a Medicare/Medicaid managed care program. The facility data reports that the median and average proportion of resident-days from such payer-types are 5.7% and 9.2% respectively. In both these cases, residents face restrictions on their choice set which we do not observe. We also don't know the prices that nursing homes get from these residents. For these reasons, we do not model demand from these residents and treat them as exogenous on the supply side.

[28] For length of stay, it would be ideal to use an anticipated length of stay instead of actual length of stay to correspond more closely with a resident's decision making. While the MDS2.0 contains an anticipated stay variable, we find that there are too many missing cases for this to be usable. We incorporate realised length of stay by bucketing stays into 3 groups. Stays within 30 days, stays from 30-100 days and stays of 100 days or over. The 30 day choice coincides with bucketing used for the anticipated length of stay in the MDS data, while the 100 day choice corresponds to the Medicare limit.

that they do not take into account how inertia could affect their future decisions. The presence of inertia could introduce invest-harvest incentives for nursing homes to price and choose quality dynamically. We are not aware of any papers or reports with these concerns, thus we abstract from these possibilities in modelling.

Preferences for quality TN_{jt} (as measured by total nursing hours per resident day) vary by whether the resident is a private, Medicare or Medicaid payer, length of stay and a large range of other demographics, including those in table 3.[29] Preferences for price p_{jt}^{priv} vary with whether a resident pays the private price on entry, their age, their length of stay, whether they have at least a bachelor's degree and whether they have a legal guardian who is not themselves. Allowing for significant heterogeneity in preferences for quality and private price supports the ability of the model to explain market segmentation through demand heterogeneity. Nursing homes may choose different private prices and qualities to avoid competing across similar pools of residents.

We also allow residents to have preferences for NFP providers. This preference captures the idea of NFPs as solving information asymmetry problems. If quality is unobserved or hard to verify, residents may prefer NFP providers as their incentives may be more closely aligned with their own residents' (Weisbrod, 1988). Although residents in our model observe a nursing home's total nursing hours per resident day, there may be other aspects of quality that a resident has imperfect information about. For example, this could include how well-trained nurses are for the nursing home. Differences in quality that are unobserved to the econometrician and systematic across NFP versus FP may also be captured in this preference. For example, our data on nursing homes does not include information about the quality of the accommodation. If NFP providers provide better quality accommodation on average than FP providers, and residents value this, it would be picked up in this preference for NFP providers. Given this interpretation of NFP preferences, we allow these preferences to vary with the same set of individual characteristics as with observed quality (total nursing hours per resident day).

λ_j are zip-fixed effects and λ_t are year fixed effects. For the estimation of mean preference parameters, we use instruments that are weighted by distances between nursing homes. These distances will be larger on average in less dense areas which could correlate with demand. Conditioning on zip fixed-effects mitigates this concern. ξ_{jt} are the unobserved demand shocks and ε_{ijt} are the standard type 1 EV idiosyncratic demand shocks. For estimation we can define nursing

[29] This includes age, a resident's depression rating scale score - a measure of mental health, cognitive performance scale score, activities of daily living score, clinically complex scale score, whether they have alzheimer's, are female, are nonwhite, have at least a bachelor's education and whether they have a legal guardian who is not themselves.

home mean utilities:

$$\delta_{jt}^{\text{priv}} = \alpha^{TN} TN_{jt} + \alpha^{p,priv} p_{jt}^{\text{priv}} + \alpha^{NFP} NFP_{jt} + \lambda_j + \lambda_t + \xi_{jt}^{\text{priv}}$$
$$\delta_{jt}^{\text{non-priv}} = \alpha^{TN} TN_{jt} + \alpha^{NFP} NFP_{jt} + \lambda_j + \lambda_t + \xi_{jt}^{\text{non-priv}}$$

For tractability, we restrict a resident's choice set to all nursing homes within 30 miles of their prior residence, as 95% of residents in the estimation sample visit nursing homes within this distance. We also make three types of exclusions for facility-years, where excluded facility-years are pooled into an outside option with utility normalised to 0.[30] First, we assume that any data are missing at random and exclude facility-years if there is a missing observable. Second, we drop a facility-year if there are less than 5 never-private payers or 5 residents who pay the private price during their stay. Finally, we drop facility-years where the proportion of Medicare bed-days served exceeds 90% following Hackmann (2019) and Feng et al. (2008) to avoid modeling resident choices where extensive and expensive rehabilitation is required at highly specialized facilities. After these exclusions, our demand estimation sample is comprised of around 940,000 nursing home stays.

Estimation and Identification

We estimate demand using a two-step procedure. In the first step, we recover $(\delta_{jt}^y, \beta_i^k, \beta_{1i}^d, \beta_{2i}^d, \beta^I)$ by maximum likelihood. As is standard in demand estimation, conditional on δ_{jt}^y, the remaining variation in price and quality is exogenous. In the second step, we recover the remaining mean preference parameters with a generalised method of moments estimator. Private prices and quality are correlated with ξ_{jt}^y thus we need to instrument for them.

We include in our instruments four types of excluded cost shifters. First, we use the distance-weighted number of FP and NFP providers in the market. The number of FP and NFP providers in the market is an instrument based on excluded product characteristics. Conditional on zip and year fixed-effects, they shift the nursing home's price and quality, but are not correlated with ξ_{jt}^y. Second, we use the distance-weighted average proportion of Medicaid and Medicare residents across other facilities. Different groups of residents have different care needs and thus, different costs. The proportion of Medicaid and Medicare residents are at the market-level, and distance between nursing home facilities is exogenous, thus conditional on zip and year fixed-effects, these instruments should not be correlated with ξ_{jt}^y. Third, we use the nursing home's Medicaid per resident-day reimbursement rate. These rates are set prospectively by state authorities, thus conditional on zip and year fixed-effects they should not be correlated with demand shocks. On the other hand, they shift a nursing home's quality choice as they affect the profitability of Medicaid

[30] Note that the universe of residents are those who choose a nursing home (but not necessarily one we observe). That is, the outside option does not include choices made by old people who choose not to go to a nursing home.

residents (Hackmann, 2019). Finally, we use the distance-weighted sum of registered nurse hours and distance-weighted sum of nursing assistant hours across other facilities. These measures capture the local availability of nurses which affects the nursing home's costs, but not the nursing home's own demand, conditional on zip and year fixed effects.

Estimation Results

Table 7 presents key parameters in our demand estimates. A full list of parameter estimates with demographic interactions can be found in Table B1 in the appendix. We further summarise the variation in estimated elasticities and estimated preferences for NFP providers in Figure 2.

Table 7: Main demand estimates

Coefficient	Estimate	Standard Error
Distance (miles)	-0.1195	(0.0009)
Distance squared	-0.0001	(0.00007)
Inertia	3.9590	(0.0037)
Private Price	-0.0248	(0.0051)
Total Nursing	-1.3257	(0.7248)
Total Nursing x Medicaid Payer	-0.3370	(0.0063)
Total Nursing x Medicare Payer	0.6284	(0.0041)
Total Nursing x Age	0.0235	(0.0003)
NFP	-4.0118	(0.2112)
NFP x Medicaid Payer	-0.5269	(0.0109)
NFP x Medicare Payer	0.8267	(0.0049)
NFP x Age	0.0514	(0.0008)

Note: These results use individual-level data from 2000-2010 in California. First stage parameters are estimated by maximum likelihood using a choice panel with around 77 million nursing home - individual observations. Second stage parameters (baseline preference for private price, total nursing and NFP) are estimated by TSLS with year and zip fixed effects with 12,990 pays private-price/never pays private price - nursing-home - year mean utilities. F-statistics for the first stage of the TSLS are 116.06 for total nursing hours per resident day and 20.10 for private price. Standard errors computed by block (drawing facilities) bootstrap with 400 iterations.

Our distance parameter estimates suggest that residents dislike being at a nursing home far away from their prior residence.[31] Figure 2 also shows many residents who are more sensitive to

[31]The average distance elasticity is -1.31 which is considerably smaller than those found in Gandhi (2023), who also

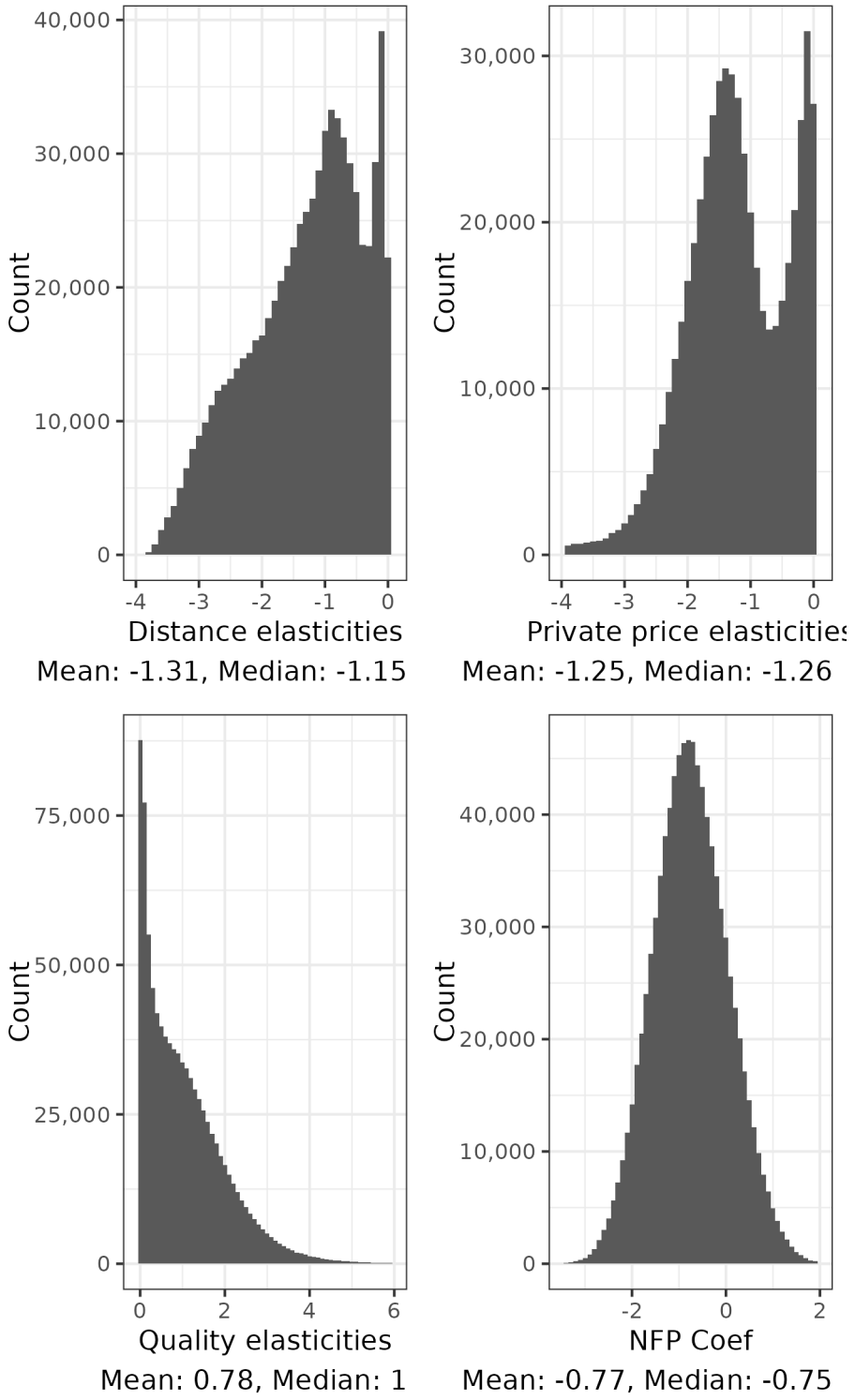


Figure 2: Individual demand elasticities

distance than the average resident. The inertia parameter estimates indicate that residents have a strong preference for staying at the same nursing home if they have multiple nursing home stays. This is consistent with the substantial disruptions to medical care and relationships with staff that a resident would experience in switching nursing homes. It is worth highlighting that the relative size of this parameter means that inertia also plays a substantive role in the counterfactuals. In particular, this implies that following a takeover, the provider has some latitude in adjusting price and quality, as inertia dampens the need to compete.

Residents dislike the private price if they have to pay. The average price elasticity is -1.25. In this case, the average is somewhat misleading as the histogram shows significant masses of residents with elasticities on either side of -1.25.[32] As shown in the appendix, we find that residents who enter the nursing home as a private payer are more price sensitive, consistent with the immediate saliency of private prices. We also find that residents who have at least a bachelor's degree are less price sensitive.

Residents value quality (total nursing hours per resident day). The histogram shows that preferences for quality are almost entirely positive.[33] The negative baseline parameter estimate likely arises because of the quality and age interaction.[34] On average, we obtain a quality elasticity of 0.78 which is comparable to those in Gandhi (2023). Again there is significant variation in the quality elasticities. Roughly half of the quality elasticities are inelastic indicating a large proportion of residents with weaker preferences for quality. On the other hand there is also a large tail of residents with stronger preferences for quality. Two systematic patterns we highlight here are that those who are a Medicaid payer at the time of entering a nursing home have a lower preference for quality compared to their private payer, and especially, Medicare counterparts. This is consistent with the observed market segmentation, where NFP providers serve more private payers and higher quality, while FP providers serve more Medicaid payers and lower quality. In the appendix, we find that residents with higher medical and nursing needs such as those with higher ADLs and CLNs, and those who are at least bachelor educated have stronger quality elasticities, on the other hand those with weaker decision making capabilities such as those who are more depressed, cognitively challenged, with Alzheimer's or who have someone else making decisions have weaker preferences for quality.

studies the Californian market like we do. This is likely because we truncate resident choice sets at 30 miles (about 48.2 km) rather than 20 kilometers, thus our data contains a stronger tail of residents who travel further than the data used in Gandhi (2023).

[32]Here, we ignore residents who never pay the private price. The price elasticity histogram is truncated on the left at -4. Around 0.6% of stays have a price elasticity below -4.

[33]The quality elasticity histogram is truncated at the left at 0. Around 5% of resident-stays have a negative quality elasticity

[34]We include age as a continuous variable to reduce parameters. Since we restrict our sample to residents aged 65 and above, it is more useful to think of the baseline parameter estimate as -1.3257 + 65$\times$ 0.0235 = 0.2018

On average, there is a bias against NFP nursing homes, although there is still a considerable tail of residents who prefer NFP providers. This could reflect two things. First, conditional on nursing homes with the same total nursing hours, residents do not view NFP status as a credible signal of nonpecuniary motive and they are not willing to pay for any additional amenities which a NFP provider might provide. Second, we do not model capacity constraints and provider selection of residents which can be important on the margin as in Gandhi (2023).[35] NFP providers are smaller on average compared to FP providers, thus residents may be less likely admitted into a NFP nursing home on average. Overall, we think it's reasonable to interpret much of these estimates as reflecting preferences. Residents with the longest stays are more likely to prefer NFP providers, which as Gandhi finds are the most likely to be rejected for admission. We also find other systematic shifters of preferences for NFP providers that are reasonably similar to systematic shifters of preferences for quality such as those with at least a bachelor's education being more likely to prefer NFP providers. A reasonable interpretation of these results is that residents who have a more 'salient' nursing home stay will care more about dimensions of quality that are harder to observe such as accommodation and food quality and therefore value NFP providers who on average do better on these dimensions of quality. This interpretation concords with our suggestive evidence discussed with our supply results, that NFP providers spend more on these other dimensions of quality.

5.2 Supply

As noted earlier, in each market, providers first observe their unobserved demand and cost shocks and then compete by choosing their private (per-day) price and total nursing hours in a static oligopoly.

The nursing home's problem is

$$\max_{TN_{jt}, p_{jt}^{priv}} \underbrace{\sum_{\tau=\text{mcaid, mcare, priv,ins}} q_{jt}^{\tau} p_{jt}^{\tau}}_{\text{Total Revenue}} - \underbrace{C^{f}(q_{jt}^{\tau}, TN_{jt})}_{\text{Total Costs}} + \underbrace{\alpha_{Q}^{f} Q_{jt}}_{\text{nonpecuniary motives}}$$

where $q_{jt}^{\tau} := \sum_i s_{ijt}^{\tau} Days_i^{\tau}$ is the quantity of resident-days for each payer-type served and $Q_{jt} = \sum_{\tau} q_{jt}^{\tau}$ is the total quantity of resident-days served across all payer-types and $f \in \{FP, NFP\}$.[36]

[35] Gandhi (2023) estimates a structural model of admissions to California skilled nursing facilities to study selective admission practices. He finds evidence that facilities disproportionately exclude Medicaid-eligible patients with long anticipated stays. Still, he finds that even nursing homes close to full occupancy, admit Medicaid-eligible patients with around 70% probability. Thus we think it's reasonable to abstract from selective admissions and focus on pricing and quality choices instead for this paper.

[36] Note that for the empirical model, we effectively use facility-reported payer-type quantities, which do not necessarily match the quantities aggregated from our demand data. This can arise for various reasons such as residents with missing observables.

Providers choose their total nursing hours and private prices to maximise an objective which depends on both the profits they make and the quantity-of-care they provide. Total revenues are the aggregate of revenues from each payer-type.[37] α_Q^f governs how much providers of a particular organisation type value providing quantity in equivalent profits and are parameters to be identified and estimated. They vary by f meaning that a separate parameter for each organisation type will be estimated. Although the nonpecuniary motive directly places value on the quantity of care provided, it can also incentivise higher quality because residents value quality when choosing nursing homes.[38]

Firm cost functions are as follows:

$$C^f(q_{jt}^\tau, TN_{jt}) = \underbrace{Q_{jt}(\gamma_0^f TN_{jt} + \gamma_1^f TN_{jt}^2 + \gamma_2^f K_{jt} + \gamma_3^f K_{jt}^2 + \mu_j + \mu_t + \eta_{jt})}_{\text{Costs common across payer-types}} + \underbrace{\sum_{\tau=\text{mcaid,mcare}} \phi^{\tau,f} q_{jt}^\tau}_{\text{Costs arising from payer-types}} + \underbrace{U_{jt} TN_{jt}}_{\substack{\text{Unobserved} \\ \text{quality fixed costs}}}$$

where cost functions differ systematically across firm types f, which allows us to explore differences in firm efficiencies.

We highlight three components of costs. First, nursing homes incur common costs across payer-types, including a common total nursing hours cost, and the variable costs of operating the facility captured by K_{jt} the size of the facility (allowing for economies of scale) and zip and year fixed effects μ_j, μ_t. Crucially, we allow for the common total nursing hours cost to have curvature, through γ_1^f. If $\gamma_1^f > 0$, nursing homes face additional cost pressure in raising quality, reducing the incentive to offset higher prices with higher quality. This is important for the existence of well-behaved equilibria. Finally, η_{jt} captures unobserved facility marginal cost shocks.

The second component of costs allows nursing home costs to vary with the distribution of resi-

[37] 'priv' and 'ins' refer separately to payer-days where the resident pays the private price and 'ins' when the resident's stay is covered by long-term insurance. As long-term insurance makes up a small proportion of stays - 6% - we do not model these distinctly and will think of such stays as being funded by a program like Medicare or Medicaid where the resident does not pay the private price. Analogously, in demand estimation we treat a resident who is covered by long-term insurance as not paying the private price.

[38] An alternative approach would be to follow Newhouse (1970) who describes a model of NFP providers who have both quality and quantity-of-care non-pecuniary motives. Since there are already many supply parameters to be estimated and separately identifying two distinct conduct parameters is especially challenging, we opt for parsimony and focus on a quantity-of-care motive only. Other papers often capture other motives by assuming that firms care about consumer surplus in addition to profits. We decided against using consumer surplus for two reasons. First, using consumer surplus also implies that firms consider willingness to pay in their nonpecuniary motive. We think it is more plausible that nursing homes, particularly NFP nursing homes, would simply weight residents equally in their nonpecuniary motive. Secondly, the construction and interpretation of consumer surplus for both the econometrician and nursing home is not as straightforward as not all residents face the private price.

dents across payer-types. As nursing homes serve more Medicaid and Medicare patients they may incur higher or lower marginal costs. This specification of costs allows us to tractably capture payer-type margins that vary systematically across firm type. The concept accords with our interviews with industry where it was common to see them discuss the 'margin' on each payer-type.

Finally, the last term captures unobserved costs only associated with quality. The latter is needed to rationalise a firm's quality choices with the data through the quality first-order condition, as in Fan (2013).

Estimation and Identification

For identification, it is useful to consider the first-order conditions abstracting away from the cost function:

$$[p_{jt}^{priv}]: \frac{\partial}{\partial p_{jt}^{priv}}\left(\sum_{\tau} q_{jt}^{\tau} p_{jt}^{\tau}\right) = -\alpha_Q^f \frac{\partial Q_{jt}}{\partial p_{jt}^{priv}} + \frac{\partial C^f\left(q_{jt}^{\tau}, \eta_{jt}\right)}{\partial p_{jt}^{priv}}$$

$$[TN_{jt}]: \frac{\partial}{\partial TN_{jt}}\left(\sum_{\tau} q_{jt}^{\tau} p_{jt}^{\tau}\right) = -\alpha_Q^f \frac{\partial Q_{jt}}{\partial TN_{jt}} + \frac{\partial C^f\left(q_{jt}^{\tau}, \eta_{jt}, U_{jt}\right)}{\partial TN_{jt}}$$

The left-hand side of these first-order conditions are the marginal revenues from adjusting price or quality and are known given our demand system.[39] As usual, we can invert these equations, project them onto the observables and recover parameters with assumptions on the unobserved cost shocks, η_{jt} and U_{jt}.

This representation highlights that α_Q^f can be viewed as rotating marginal revenues outwards. When $\alpha_Q^f > 0$, firms have nonpecuniary incentives for serving a higher quantity and are willing to incur a higher level of marginal cost. Therefore, the identification challenge with α_Q^f is to distinguish between higher levels of marginal cost arising from firms with more inefficient marginal cost function parameters, or firms which care more about providing quantity-of-care.[40]

We follow a similar idea as in Hackmann (2019) and Kim (2024) and match the mean of simulated variable costs with the mean of actual variable costs:[41]

$$\mathbb{E}\left[VC_{jt} - \hat{VC}_{jt}(\theta_s^f)\right] = 0,$$

[39] Note that in general, $\frac{\partial q_{jt}^{\tau}}{\partial p_{jt}^{priv}}, \tau = \{mcaid, mcare\}$ need not be 0 due to residents who transition to and from paying the private price.

[40] Specifically, since α_Q^f is a level shifter of marginal costs we cannot distinguish it from μ_j, μ_t without an additional source of variation.

[41] Constructing variable costs from the data requires taking a stance on what is 'variable'. We use the categories of operating expenses by cost center in the cost report data adding the following categories: Skilled Nursing Care, all Ancillary services except Home Health Services and all support services except Administration. We do not include property expenses, interest expenses and provisions for bad debts which are likely to be fixed costs.

where VC_{jt}^f are variable costs from cost report data, and $\hat{VC}_{jt}^f$ are simulated variable costs. [42] We match variable costs because unlike those two papers, our cost function exhibits nonlinearities from $\phi^{\tau,f}$. Thus, we cannot compute marginal costs from data (as say, average variable cost) and directly match marginal costs. Cost-function parameters govern how variable costs grow with quantity, whereas the nonpecuniary motive α_Q^f does not enter variable costs directly. Thus, conditional on observed quantities and quality, differences in reported variable costs discipline differences in cost-function parameters. These cost moments allow us to distinguish genuinely higher marginal costs from a greater willingness to incur marginal costs arising from the quantity-of-care motive.

Some care is needed in constructing cost moments. Gandhi and Olenski (2024) show that nursing homes – especially FP providers – often engage in the tunneling of profits through the use of related parties, meaning providers could overstate costs in their cost reports. An owner of a nursing home may own or have financial interests in several other businesses which are not directly vertically integrated and operate independently. For instance, in addition to the nursing home, they could have financial interests in businesses operating in catering, cleaning, medical goods, real estate and so on.[43] The nursing home could contract with these related party businesses at inflated prices and through Medicaid and Medicare reimbursements which partially scale with past reported costs, extract rents from Government.

We address this concern in two ways. First, Gandhi and Olenski (2024) show that this practice occurs primarily through property costs and to a lesser extent management costs. In our cost functions these are thought of as fixed so we do not match on these. Still, our construction of variable costs is based on cost center definitions and therefore quite broad, so it may be hard to remove all of the fixed cost components. Therefore, we introduce an additional cost moment, where we match on the within-facility first-difference in variable costs:

$$\mathbb{E}\left[(VC_{j,t} - VC_{j,t-1}) - \left(\hat{VC}_{j,t}(\theta_s^f) - \hat{VC}_{j,t-1}(\theta_s^f)\right)\right] = 0,$$

First-differencing variable costs removes level (fixed) components. Thus, rather than relying on cross-sectional cost levels, this moment primarily draws on within-nursing-home changes in variable costs as the quantity and composition of patient-days served change over time.

The remaining identification challenges are standard. Since the firm chooses TN_{jt} and p_{jt}^{priv} knowing η_{jt} and U_{jt}, they are likely correlated. Thus we need instruments Z_{jt} which shift TN_{jt} and p_{jt}^{priv} but are not correlated with cost otherwise.[44] With instruments Z_{jt} in hand, our moment

[42] In constructing this moment we winsorize the data at the 98th percentile for average variable costs. This avoids matching on abnormally high variable cost reports. In addition, since quality fixed costs are fixed from the perspective of quantity they are also not included in the matching of variable costs.

[43] See Rau (2017) (https://kffhealthnews.org/news/care-suffers-as-more-nursing-homes-feed-money-into-corporate-webs/) for an interesting investigative piece that provides examples of related party usage.

[44] We treat K_{jt} as exogenous. In the data we observe very few changes in capacity over time. Across all providers and

conditions are:

$$\mathbb{E}\left[Z_{jt}\eta_{jt}\right] = 0,\ \mathbb{E}\left[Z_{jt}U_{jt}\right] = 0$$

We estimate supply side parameters by GMM using the moments above, separately for FP versus NFP providers.[45] This means we do not impose any further parametric structure on how costs between NFP and FP providers can differ.

Our rich cost function specification has many endogenous variables for each moment condition. Accordingly, many instruments are required. The first set of instruments are those relating to competitors. The number of distance-weighted FP and NFP nursing homes, the distance-weighted sum of total beds and distance to nearest nursing home. All of these relate to exogenous characteristics about competitors which are not correlated with the nursing home's own η_{jt}, U_{jt}. These should shift the quantities that the nursing home faces and the quality they choose in response to those exogenous factors.

The second set of instruments relates to discharges to nursing homes from other medical facilities. Discharges to nursing homes from other types of medical facilities are unlikely to be correlated with the nursing home's own η_{jt}, U_{jt}, but shift the quantities they are likely to serve. We use the distance of a nursing home to the nearest hospital, as hospitals can discharge residents directly to nursing homes. We use distance-weighted hospital admissions to proxy for discharges from hospitals to nursing homes.

In addition to hospital discharges to nursing homes we also draw on home-health discharges. Home-health services provide nursing care to residents in their own home instead of at a nursing facility. Since nurses cannot typically be at someone's house 24/7, people who find themselves needing more intensive supervision can be discharged to a nursing facility. We have home health data that provides the total number of discharges to intermediate care facilities or skilled nursing facilities at a facility-year level. This enables us to construct distance-weighted discharges for a particular nursing home. The data we use also breaks down home health quantities by payer-type (private-payer, Medicare, Medicaid), but not specifically the number of discharges to particular nursing homes. Since this directly speaks to our payer-type quantities we also include these distance weighted home-health quantities. All else equal a higher quantity served at a home health facility also results in more discharges.

Finally, we also use lagged total nursing home penalties as an instrument. Nursing homes typically undergo annual inspections, which are administered by state survey agencies and required by CMS. These inspections assess compliance with federal health and safety requirements and

years we only see expansions of capacity in 3.4% of cases and contractions of capacity in 4.4% of cases.

[45] In constructing the sample moment conditions, it is worth noting that U_{jt} does not show up in the pricing first order condition. Thus for a given guess of parameters, we can compute η_{jt} and then use the quality first order condition to back out the implied U_{jt}

cite deficiencies when facilities fall short. When deficiencies are sufficiently serious or unresolved, regulators may impose penalties on providers. Facilities are required to correct the cited deficiencies and demonstrate a return to compliance. The requirement to resolve deficiencies means that lagged penalties are likely to shift contemporaneous quality, while being uncorrelated to contemporaneous costs. Indeed, Gandhi et al. (2025) demonstrate that nursing homes and their 'effort' are responsive to the potential for an inspection, and following an inspection due to revisit inspections for deficiencies.

Estimation Results

Table 8: Supply Parameters

	NFP		**FP**	
	Coefficient	Standard Error	Coefficient	Standard Error
Panel A: Non-Pecuniary Objective				
Quantity Motive (α_Q^f)	32.98	(13.11)	-2.78	(14.24)
Panel B: Cost Function Parameters				
Quality Cost (γ_0^f)	-13.11	(19.73)	-27.22	(17.15)
Quality Cost Sq (γ_1^f)	16.38	(8.14)	17.33	(8.69)
Mcaid Quantity ($\phi^{mcaid,f}$)	108.67	(57.00)	-16.43	(47.32)
Mcare Quantity ($\phi^{mcare,f}$)	87.70	(202.12)	9.19	(245.78)
Facility Size (γ_2^f in 100 beds)	86.76	(103.19)	-36.88	(20.39)
Facility Size Sq (γ_3^f in 100 beds)	-62.82	(54.84)	10.43	(6.05)
N	632		4857	

Note: Parameters can be interpreted in $/bed-day. Zip and year fixed effects included. These results use facility-level data from 2000-2010 in California. Standard errors computed using the delta method, combining the supply-side GMM (clustered at the firm level) and the block bootstrap using 400 iterations from the demand-side. 1st stage F-statistics for each of the endogenous variables are reported in Table B3.

We begin with our supply parameter estimates in Table 8. All of these can be interpreted in terms of dollars per bed-day. The objective parameter estimates indicate that NFP and FP providers value providing an additional bed-day of care at approximately $33 and -$3 per bed-day respectively. These results are consistent with a broader literature that finds NFP nursing homes exhibit some form of altruistic motive and confirm common perceptions that NFP providers have non-pecuniary motives. They are also consistent with regulations requiring NFP providers to provide public ben-

efits in return for their generous tax treatment.[46] To gauge the relevance of the $33 estimate, note that the average NFP private price is approximately $198 per bed-day so this represents about 17% of the price or 21% of their average margin which is $158 per bed-day.

Turning to quality costs, since we estimate positive and significant convex terms, quality becomes increasingly costly for both FP and NFP providers. While the linear terms are insignificant, FP providers have a larger negative component than NFP providers which will lead to lower quality costs in the model. This is consistent with our motivating evidence where we saw that NFP providers use a significantly higher skilled mix of nursing hours compared to FP providers (Table 4), which can lead to higher quality costs.

Table 9: Firm-level payer-type marginal costs

	Mean	25 Pct	Median	75 Pct
Panel A: Private				
FP	91.9	79.9	112.1	142.4
NFP	40.3	43.2	85.3	118.9
Panel B: Medicaid				
FP	75.5	63.5	95.7	126.0
NFP	149.0	151.9	193.9	227.6
Panel C: Medicare				
FP	101.1	89.1	121.3	151.6
NFP	128.0	130.9	173.0	206.6

Note: All costs in $/bed-day.

To best understand how costs vary across types of providers and payer-types, Table 9 calculates model implied marginal costs. We highlight four patterns in the estimated marginal costs arising from the demand side – resident characteristics – and the supply side – cost report data that could explain variation in estimated marginal costs. Resident characteristics affect cost estimates through inversion of the FOC, while cost report data also directly affect cost estimates through the cost matching moments.

First, marginal cost for Medicaid varies significantly across FP and NFP providers. For FP providers, Medicaid payers are the least costly to serve with private and Medicare payers being more costly. This is consistent with the price hierarchy as shown in Table 2. For NFP providers,

[46]The result for NFP providers appears to be of a similar magnitude as in Hackmann (2019). Hackmann (2019) considers government and private NFP nursing homes separately and estimates that they act as if they have a marginal cost advantage of $36 and $23 respectively.

Medicaid residents are the most costly to serve, consistent with such payers being older and having especially long stays on average (Table 3). We interpret marginal costs as a discounted average per bed-day marginal cost meaning that these will be larger as costs grow over the duration of a resident's stay. A body of literature suggests that long stays may incur increasing costs in latter stages of the stay as residents can progressively lose function and become more dependent on nursing and medical care. A number of studies which follow nursing home residents observe increases in resident ADL scores over time which are hard to reverse (Talley et al., 2015; McConnell et al., 2002; Chen et al., 2007). In addition to physical decline, the probability of hospitalizations increases with length of stay and Kruse et al. (2013) find that hospitalizations also lead to significant increases in ADL scores. This even applies for the common flu Gozalo et al. (2012).

Second, FP providers have a significant cost advantage in serving Medicaid residents, with a gap compared to NFP providers of on average around $75 per bed day. Although Medicaid payers also have long stays compared to Private and Medicare payers at a FP nursing home, the extent to which these matter for costs also depends on how the nursing home operates. In appendix B.3, we study differences in cost center spending across FP versus NFP providers and observe that NFP spend around $20 more per bed-day due to differences in the usage of nursing assistants and differences in non-medical care such as accommodation and food. We interpret these differences to represent real differences in dimensions of quality other than licensed nurses as opposed to differences in operational efficiency. First, our demand estimates suggest that the residents most likely to value non-medical quality have stronger preferences for NFP providers – which could reflect systematic differences in unobserved quality. This includes those with especially long stays, who have higher degrees and those who are responsible for their own care decisions. Another reason we find this plausible is that it is consistent with the NFP business model of operating upstream facilities (independent living and assisted living facilities) whose residents are more likely to pay a private price. Quality should be especially important for those facilities for attracting residents who have more outside options, for example staying at home.

Still, cost center data only allows us to observe the total spend as opposed to input choice. We would also expect some of the total $75 gap to be explained by differences in operational efficiencies such as inefficient contracting or usage. Lu and Lu (2021) study conversions from NFP to FP nursing homes and find that converted FP nursing homes are able to reduce costs by reducing nursing administration costs which is consistent with some FP providers having more efficient cost functions.

Third, for both FP and NFP providers, Medicare costs are relatively expensive compared to private payers because Medicare stays usually follow hospitalization and require intensive rehabilitation therapies which can lead to a relatively high cost of care. This is consistent with table B4 (in appendix B.3) which shows that ancillary costs (which includes therapists) dramatically raise

nursing home average costs.

Fourth, the relatively low cost of NFP providers for serving private payers is also noticeable. This leads to NFP providers having a cost advantage of around $50 per bed-day on average compared to FP providers when serving private payers. We think this reflects two factors. First, NFP providers tend to serve relatively more price inelastic residents, including those with long-term care insurance. Holding prices fixed, a more inelastic price demand implies lower marginal costs. The average individual choice elasticity with respect to private prices of residents at NFP providers is -0.66 versus -0.74 for FP providers. Second, as we discussed in section 2.3, NFP providers tend to operate upstream facilities such as assisted living and independent living facilities. Since assisted living and independent living facilities are primarily paid by private means and tend to provide more non-medical care, NFP nursing homes may experience economies of scope and share costs with these facilities leading to lower private-payer costs.[47]

6 Explaining Quality and Access Patterns

With our estimated model of nursing home demand and supply, we revisit and explain our motivating evidence through the lens of the model. Recall that NFP providers are more prevalent in higher-income markets and serve more private-pay residents, while FP providers serve more Medicaid residents, who are more prevalent in lower-income markets. We hypothesised that these patterns arise because NFP providers find private-pay residents relatively more profitable, while FP providers are better positioned to serve Medicaid residents. Our estimated model allows us to examine this hypothesis using payer-type margins, which are not directly observed.

Focusing on figure 3, the rightwards shift in the distribution of Medicaid margins for FP relative to NFP providers is a clear indication that FP providers find Medicaid residents profitable compared to NFP providers.[48] On the NFP side, Medicaid margins appear to be the worst on average with the private margin being largest. This is consistent with the observed patterns of access. The fact that a large proportion of Medicaid margins is negative is also consistent with industry reporting. For example, the American Health Care Association reports that two-thirds of surveyed nursing homes are reimbursed for less than 80% of the cost of providing care for Medicaid payers and only 10% report reimbursements which cover 90-100% of the cost of providing care for Medicaid payers (American Health Care Association, 2025).

[47] Again, we cannot address these upstream facilities in our supply model since there is no readily available data about these upstream facilities.

[48] All figures are truncated at -100 on the left and 300 on the right. For private margins 0.5% of firms have a margin below -100 and 3.1% firms have a margin above 300. These numbers for Medicaid margins are 1.9% and 2.7% respectively and for Medicare margins 0.1% and 3.6% respectively.

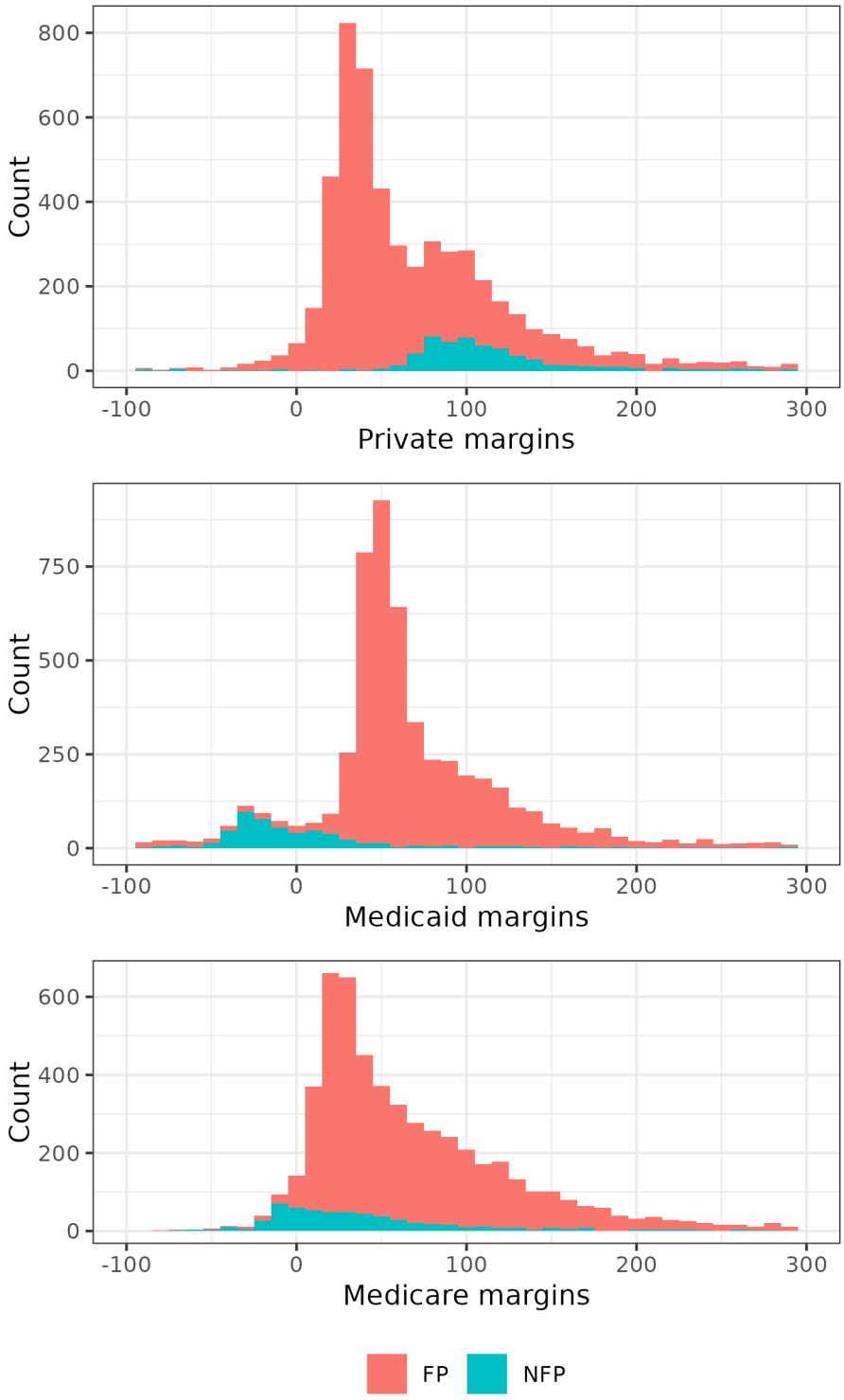


Figure 3: Payer-type margins, at observed quantities

Next, we use counterfactuals to examine how differences in costs and non-pecuniary motives explain NFP providers' higher quality choices and market segmentation, accounting for equilibrium responses. For tractability, all counterfactuals in the paper focus on 25 representative small markets – MSA-years with less than 20 nursing homes that are on average similar in market size, share of FP providers and payer mix.

We consider changes to three parameters: non-pecuniary objectives, marginal costs, and Medicaid quantity costs. First, a NFP provider's nonpecuniary motive for quantity-of-care could incentivize them to choose higher quality to serve more residents. Second, NFP providers may prefer to avoid directly competing with FP providers and their cost advantage in serving Medicaid residents, by serving residents with stronger preferences for quality. Thus, differences in marginal costs could also lead to differences in quality choices. Finally, unlike changes in general marginal costs, changes in Medicaid quantity costs directly affect the relative cost of serving different payer types. Given Medicaid's large share of bed-days these changes may have distinct effects on quality choices.

Figure 4 presents the results from these counterfactuals. In each plot, we present the average price and quality choice across FP and NFP providers in all small markets under different choices of NFP supply parameters. For the NFP objective parameter we consider changes ranging up to \$35 per bed-day which includes the scenario where the NFP objective is completely removed. For changes in marginal costs, we consider changes ranging up to \$75 per bed-day which includes scenarios where NFP and FP providers have the same average marginal cost for serving Medicaid residents.[49] We add arrows and labels to indicate the direction in which average prices and quality choices change as the magnitude of the change in NFP supply parameters changes.

In Panel A, we study how price and quality choices adjust as the NFP quantity motive is reduced. Weakening the NFP motive leads NFP providers to derive less utility from providing more bed-days of care, so they increase their prices and reduce quality – enough to reverse patterns of quality across FP and NFP providers – as they orient towards a purely profit-driven motive. The extent to which the adjustment occurs through price or quality is ambiguous and depends on the distribution of preferences for price and quality. In this case, increases in private price dominate reflecting that NFP providers serve more price inelastic residents and residents with stronger preferences for quality. As NFP providers reduce quality and increase their private prices, FP providers increase their quality (and price), to compete for residents with stronger preferences for quality. These counterfactuals suggest that the NFP objective is a major driver of differences in quality.

[49] Specifically, in the reduce NFP objective counterfactuals we plot the baseline and average private prices and quality under reductions in the NFP objective by 5,10,15,20,25,30 and 35. In the reduce marginal cost and Medicaid quantity cost counterfactuals we consider reductions in NFP overall marginal costs and Medicaid quantity cost by 7.5,15,22.5,30,37.5,45,52.5,60,67.5 and 75.

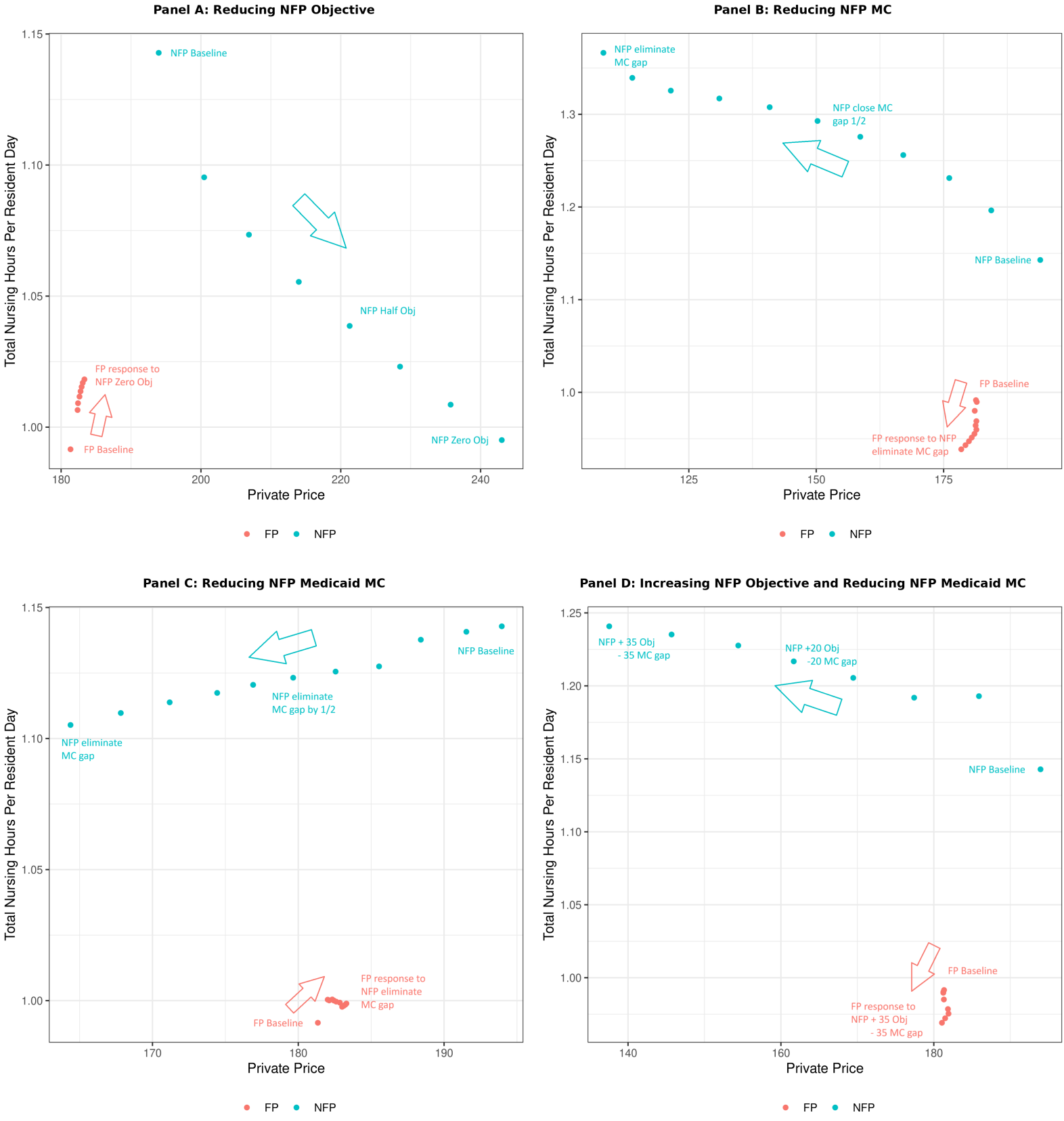


Figure 4: Counterfactuals to explain quality choices

In Panel B, we study how price and quality choices adjust as the NFP marginal costs decrease. Reducing NFP marginal costs by $75 eliminates the average marginal cost gaps on serving Medicaid payers, leads to a cost advantage for serving Medicare payers and exacerbates the cost advantage that NFP providers have in serving private payers. Accordingly, NFP providers significantly reduce their private prices allowing them to serve residents with more elastic private price demand, including those who subsequently become Medicaid residents. They also pass through the reduction in cost as higher quality, allowing them to gain market share across all payer-types. As NFP providers cut prices and increase quality, FP providers respond by decreasing their price and quality to focus on residents who are price-elastic. Since the quality gap between FP and NFP providers persists in these counterfactuals, the overall level of marginal costs is unlikely to explain differences in quality.

In Panel C, counterfactuals studying reductions in Medicaid quantity costs highlight how differences across payer types drive pricing and quality choices. While reductions in overall NFP marginal costs increased NFP quality, now NFP providers reduce quality and prices as the gap in Medicaid quantity costs is eliminated. As Medicaid residents become relatively more profitable compared to other payer types and Medicaid payers have weaker preferences for quality, NFP providers find it more valuable to reduce quality. In contrast, FP providers slightly increase their prices and quality. Now NFP providers prefer leaning into the lower end of the market, thus FP providers find it more valuable to serve residents with stronger preferences for quality. This set of counterfactuals highlights that eliminating the FP cost advantage in serving Medicaid residents is also likely a driver of differences in quality.

While these exercises are helpful in showing that both non-pecuniary objectives and Medicaid costs matter, they do not speak to their relative importance. In Panel D, we pit these forces against each other by increasing the NFP objective and reducing Medicaid costs by equal amounts. NFP quality rises while prices decline sharply. FP providers slightly reduce quality.[50] Thus, over the range considered, non-pecuniary motives have a larger effect on equilibrium quality than Medicaid costs at the existing NFP margin. Importantly however, these results concern existing NFP providers, which are better positioned to serve private-pay residents. In contrast, our takeover counterfactuals involve replacing FP providers oriented toward Medicaid residents with NFP providers. Accordingly, we expect differences in cost to matter more for the takeover counterfactual.

[50] We consider seven scenarios in each market, increasing the NFP objective and reducing Medicaid costs by 5, 10, 15, 20, 25, 30, and 35. Note that we recast the Medicaid cost grid to match the NFP objective grid so that we are comparing the same changes in levels.

7 Banning for-profit providers

Finally, we use the estimated structural model to simulate the effects of counterfactual bans on for-profit providers, motivated by policies limiting for-profit provision in certain states. Importantly, we allow for subsequent takeovers by NFP providers instead of new entry, as we observe limited entry (45 throughout our California sample) to the industry but more takeovers by FP providers (264).[51]

In each market, the lowest quality FP nursing home is taken over by a local NFP nursing home.[52] We change parameters on the demand and supply side for the targeted nursing homes and then solve for the new equilibrium. On the demand side, we add the NFP product attribute to targeted nursing homes. On the supply side, we change FP parameters to be closer to their NFP counterparts and we shift the unobserved cost shocks to be closer to the mean of the corresponding NFP cost shocks in the market. The interpretation is that these changes represent a counterfactual where the new NFP provider chooses their private price and quality, and extends their 'local' business model - incurring increases in costs that partially reflect differences in unobserved quality. Residents internalize these changes or changes in the provider's reputation due to their new NFP status when making their nursing home choice. Importantly, the direct demand effect of this change is not uniform: preferences for the NFP attribute are heterogeneous across residents, so its effect varies with the composition of each market.

We do not explicitly model a takeover process and any transition costs incurred. To that extent, we regard our evaluation as generous to the policy. Given the static nature of our model and counterfactual, we also interpret the results from a long-run point of view.

We consider three different parameterizations of the counterfactuals that reflect how much the previously FP provider changes. A 'minimal change' parameterization where all parameters of the FP provider being taken over are 1/20th a step closer to NFP parameters and we add 1/20th of the NFP parameters in demand, a 'Half NFP' parameterization where the parameters become halfway between a FP and a NFP's parameters and we add 1/2 of the NFP parameters in demand and a 'Full NFP' parameterization where the FP parameters are replaced with their NFP counterparts and the NFP parameters in demand are fully added. These parameterizations allow for the provider which is taken over to operate not entirely as a NFP provider which may be best for the population that used to be served by a FP provider.

[51] See Table A3. The observed takeovers are not informative for our counterfactuals since they are almost entirely FP to FP and likely to be selected.

[52] Since our pool of NFP nursing homes includes government providers (around 20% of NFP providers), this could also be interpreted as a government takeover.

Table 10: Aggregate Market Effects from Takeovers

	Mean	25th Pct	Median	75th Pct
Panel A: Minimal Change				
CS ($m)	-0.19	-0.27	-0.11	0.23
Total Quantity (bed-days)	3264.6	-3.7	-0.5	$\sim$0
Panel B: Half NFP				
CS ($m)	-1.39	-2.10	-0.84	-0.25
Total Quantity (bed-days)	2632.5	-14.3	-1.7	0.2
Panel C: Full NFP				
CS ($m)	-2.32	-3.74	-1.11	-0.51
Total Quantity (bed-days)	2441.8	-27.0	-3.5	1.1

Note: This table summarizes the market-level impacts of the counterfactuals under different parameterizations of the takeover. Each statistic is taken from the distribution of outcomes across markets where in each market, the lowest quality FP is taken over by a NFP provider. Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. Under the minimal change parameterization, all parameters of the FP provider being taken over are 1/20th a step closer to the NFP parameters and 1/20th of the NFP demand parameters are added. In the Half NFP parameterization, all parameters of the FP provider being taken over are 1/2 between FP versus NFP parameters and 1/2 of the NFP demand parameters are added. In the Full NFP parameterization, all parameters of the FP provider are changed to NFP parameters and full demand parameters are added.

Table 10 presents aggregate market effects under each of the three parameterizations. Across all parameterizations, we find significant losses in consumer surplus. In the 'Full NFP' parameterization we find reductions in consumer surplus of around $1.11 million and $2.32 million in the median and mean market.[53] In the majority of markets, total quantity is minimally affected. We return to the positive average increase in total quantity at the end of the section. Across all parameterizations, the 25th to 75th percentile changes are within 100 bed-days which is already quite small in the context of average nursing home total quantities being in the order of 30,000 (Table 1). The minor change in total quantity served reflects inertia in resident demand. Changing nursing homes is costly and burdensome, thus even under a takeover counterfactual, residents are unlikely to switch nursing homes altogether.

Table 11 unpacks that the reduction in consumer surplus is arising from degraded services.

[53] Consumer surplus is calculated as $CS = \sum_i Days_i CS_i$, where CS_i is the usual logit logsum formula for consumer surplus. When expressing consumer surplus in dollars, we assume that residents who never pay the private price otherwise have the same price preferences as with someone who does.

Table 11: Firm Level Effects from Full NFP Takeovers

		Mean	25th Pct	Median	75th Pct
Price	Remaining FP	23.30	-8.84	2.37	17.37
	Takeover NFP	101.04	27.54	87.38	162.63
	Existing NFP	5.11	-17.48	-2.31	5.13
Quality	Remaining FP	0.007	-0.036	0.003	0.025
	Takeover NFP	-0.531	-0.592	-0.544	-0.485
	Existing NFP	0.036	0.001	0.010	0.052

Note: This table summarizes the firm-level impacts of the counterfactuals in the Full NFP takeover parameterization where all parameters of the FP provider are fully switched to their NFP counterparts. Directionally we find similar results as in the other parameterizations as shown in table B5. Each statistic is taken from the distribution of outcomes across markets where in each market, the lowest quality FP is taken over by a NFP provider. Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. In the Full NFP parameterization, all parameters of the FP provider are changed to NFP parameters and full demand parameters are added.

Here, we focus on the 'Full NFP' parameterization.[54] We focus on two impacts: the direct impact of changes to the provider being taken over and the indirect impact of other providers' responses.

Focusing on the direct impact, NFP providers that take over an FP provider raise private prices. In the mean and median scenario, private prices are raised by around $101 and $87 respectively. Part of the increase in private price reflects existing differences in private prices between FP and NFP providers in the observed equilibria of around $15 and $25 in the median and mean respectively. The remainder of the increase is the NFP provider requiring cross-subsidization from private payers for serving Medicaid payers. This increase in prices is consistent with the patterns in the data and other structural estimates. First, the average NFP provider has a negative margin on serving medicaid payers. Second, our demand estimates suggest that residents exhibit significant inertia. FP providers serve more Medicaid residents than the typical NFP provider. Inertia means that a NFP provider cannot easily attract their usual payer-mix which is more weighted to private payers than a FP provider. Accordingly, as costs of providing care increase, the NFP provider raises private prices. Significant adjustment also occurs through degraded quality. In the average takeover, the NFP provider reduces quality to reduce costs as indicated by the reduction of 0.53 nursing hours per resident-day. The combined increase in price and reduction in quality leads to large declines in consumer surplus.

[54] Results on the other parameterizations are included in Table B5 and are directionally similar.

Indirect effects are relatively small compared to the direct effects. For the remaining FP providers, the fact that the previously FP provider has become relatively uncompetitive represents an opportunity to pry away residents, which could be profitable to attract, depending on their own costs and the distribution of payer characteristics in the market. Accordingly, remaining FP changes in prices are relatively small in most markets and increases or small reductions in quality in many markets. Existing NFP providers also have incentives to compete. Attracting residents is valuable for meeting their non-pecuniary motive. This leads existing NFP providers to maintain or shade down prices and increase quality slightly.

These results suggest that in most markets, mandating the takeover of (the lowest quality) FP providers by NFP providers is not effective. Although we do not see reductions in access through people receiving less nursing home care, the channel of harm is through higher prices and degraded service as resident inertia limits substitution to other nursing homes. These results underscore the role of FP providers in serving lower-income populations at relatively low cost, even when their removal does not substantially reduce utilisation.

Given the limited policy success in this industry, one might have hoped that NFP provision could be a silver bullet. This is especially the case because it does not necessarily involve the government spending substantially more money, which history demonstrates is a difficult ask. Ultimately, these results suggest that increased NFP provision is unlikely to be a substitute for increased government funding through measures such as paying nursing homes more as in Hackmann (2019) or a stronger regulator, which was a key recommendation in National Academies of Sciences, Engineering, and Medicine (2022).

Finally, we return to the heterogeneity in market responses. Our results regarding consumer surplus are robust across our parameterizations and most markets, but it is noticeable that on average we observe increases in total quantity served which are driven by a small proportion of markets. These cases highlight that the model also allows for these mandated takeovers to have beneficial effects depending on specific market conditions. In tables B7 and B8 we present further results that examine markets where there are relatively large increases in total quantity - 100 bed-days or more.[55] First, large change markets have more bed-days served at baseline. The larger the market and the more beds served by the provider being taken over, the larger the value in competing for residents at the provider. Accordingly, indirect effects are more competitive in these markets compared to all markets. Secondly, the providers being taken over in the large change markets serve a significantly larger proportion of private payers compared to the average market. This leads existing NFP providers to be especially competitive with large increases in quality as private payers are

[55] 100 days, as the largest 75th percentile responses are in the order of up to 100. 100 bed-days represents the 88th percentile in the Full NFP parameterization. In Table B6, we confirm that losses in CS persist after removing these markets.

especially profitable for NFP providers, and private payers have stronger preferences for quality compared to Medicaid residents.

8 Conclusion

Should policymakers reduce FP provision and increase NFP provision in regulated markets such as health and education? Overall, we argue that the balance of the model and empirical evidence suggests that it does not. In some other settings such as hospitals, papers have argued that NFP providers may simply be FP providers in disguise (Dranove et al., 2017). We find significant differences between NFP and FP providers. Consistent with past work, NFP providers have nonpecuniary motives for providing quantity of care. On the other hand, NFP providers have especially higher costs for Medicaid payers who make up the largest share of bed days.

These differences between FP and NFP providers matter. While the nonpecuniary motive contributes to NFP providers choosing higher quality, their quality choices are also driven by higher costs of serving Medicaid residents, which make private payers relatively more profitable. Since private payers have stronger preferences for quality, these cost differences provide an additional incentive for NFP providers to choose higher quality. At the same time, FP providers' lower costs of serving Medicaid residents allow them to serve lower-income populations while charging lower private prices, thereby supporting access to nursing home care. Thus, banning FP nursing homes is unlikely to be a fruitful policy solution.

Finally, we began this paper with general health and education settings in mind. We leveraged the US nursing homes industry to study this question, which necessitates addressing unique institutional features. However, similar patterns can be found in other markets such as post-secondary education (Deming et al., 2012). Accordingly, this framework and the lessons learned may have broader applicability beyond the US nursing homes industry.

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Appendix

A Supplemental facts

A.1 NFP nursing homes charge higher private prices than FP nursing homes in California

Table A1: Differences in Private Prices

	Difference in Means	Regression
FP	-17.8	-15.7
		(4.92)

Note: These results use yearly facility-level data from 2000-2010 in California. Controls include facility size (in beds), aggregate activities of daily living score which measures how dependent the population is on carer support for daily needs, aggregate case mix index which measures how much medical care the population requires and proportion of Medicaid and Medicare payers in the facility. Year and zip fixed effects are included with standard errors clustered at the facility-level.

A.2 Donations are unlikely to play a role in explaining location patterns

We argue that one reason why NFP providers are relatively more prevalent in higher income markets is because they serve a larger proportion of relatively profitable private-pay residents. Another reason could be that it is easier to source donations when the local population is wealthier on average. In interviews with industry, we heard that donations do not play much of a role as the cause of nursing home care is relatively unappealing. We check this claim empirically by drawing on supplemental data from the National Center for Charitable Statistics (NCCS) who compile and clean data from NFP annual form 990 filings. This data reports the contributions which nursing homes receive each year along with usual financial data. Contributions include both donations and government funding, but are not separately split in the NCCS data. Figure A1 is a histogram of nursing home contribution rates as a proportion of total revenues separated by the income quartile of the nursing home's zip. A few nursing homes receive all of their revenue in the form of contributions. These are likely to be government-run nursing homes. On the other hand, the vast majority receive none, and zip income quartile has very little effect on this distribution. We interpret this as clear

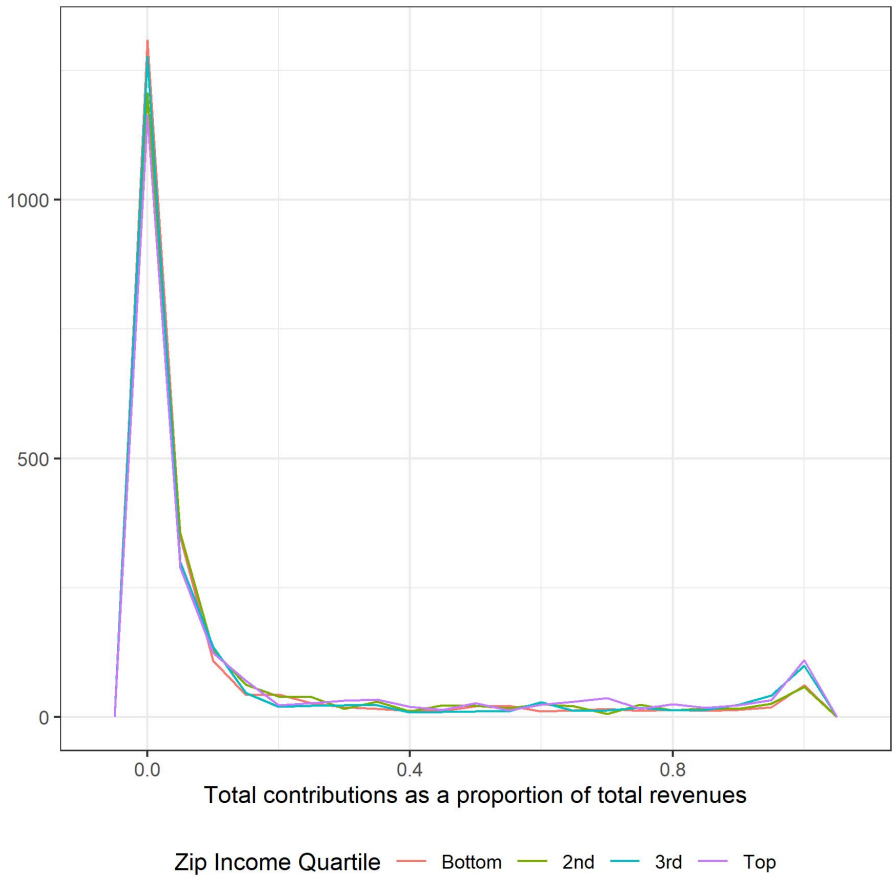


Figure A1: Histogram of contribution rates by zip income quartile, all states, 2010-2018

evidence that donations do not matter.

A.3 T-tests on Individual Characteristics

Table A2: T-tests on Individual Characteristics

Provider Type	Group Being Compared to Private	Age	LOS	ADL	CLN	CPS	Has Alzheimer's
FP	Medicaid	[0.087,0.182]	[-376.94,-371.571]	[0.664,0.709]	[0.205,0.217]	[-0.765,-0.744]	[-0.045,-0.042]
FP	Medicare	[-0.035,0.044]	[55.711,56.848]	[0.133,0.167]	[0.077,0.088]	[0.305,0.323]	[0.006,0.008]
NFP	Medicaid	[-0.426,-0.076]	[-437.439,-412.808]	[-0.059,0.116]	[0.125,0.17]	[-0.874,-0.794]	[-0.05,-0.037]
NFP	Medicare	[0.693,0.909]	[123.762,129.403]	[0.185,0.288]	[0.037,0.064]	[0.384,0.431]	[0.014,0.02]

Note: Intervals are 95% confidence intervals for the difference in means.

A.4 Entries, Exits and Takeovers in California

We focus on government-mandated takeovers for our counterfactuals. In contrast, firm entry is usually the extensive margin that papers think about. Here we demonstrate that takeovers are the more relevant extensive margin. As we saw in table 1, entry activity is limited. This suggests that the fixed costs of constructing and certifying a new facility are likely to be prohibitively large. Motivated by recent concerns about takeovers and acquisitions,[56] we consider the prevalence of takeovers. Lack of transparency in ownership is notorious in the nursing homes industry.[57] Our facility-level data assigns facility IDs based on location rather than facility-owner, thus to address this gap we undertook a hand-coding exercise to identify likely changes in ownership. This was done on the basis of changes in facility name, administrator, profit-status and googling facility histories. We summarise the results below:

Table A3: Entries, exits and takeovers in California, 2000-2010

	New facility	Facility closure	Own type takeover
FP	33	88	264
NFP	12	82	1

Note: There are very few across firm type takeovers. 17 NFP becoming FP and 2 FP to NFP.

The relative magnitude of takeover versus entry activity amongst FP providers is striking. To check that our hand-coded takeovers likely reflect true takeovers we look for evidence of changes in management with an event-study exercise.

Specifically, we estimate the following regression equation for facility j at year t:

$$y_{jt} = \alpha_j + \tau_t + \sum_{k \neq -1} \beta_k \mathbf{1}_{j,k=r(j,t)} + \epsilon_{jt}$$

where α_j are facility fixed effects, τ_t are year fixed effects, and $r(j,t)$ is the relative year from when the facility j was taken over by another facility. Figure A2 reports the estimated coefficients β_k. The sample is relatively small so our estimates are somewhat noisy. Some clear patterns emerge. Resident profit, which are revenues minus expenses for resident-care services increases substantially following a takeover. The reason appears to be that nursing homes shift away from serving Medicaid residents towards serving other resident types while keeping average variable and total costs

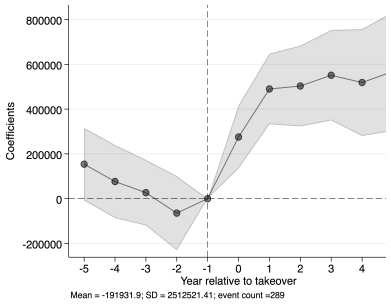
[56] Gupta et al. (2021) find that the wave of private equity takeovers increased mortality in nursing homes. Harrington et al. (2017) notes that these patterns of FP takeovers are common in many countries and are often associated with quality deficiencies.

[57] This has led to recent improvements in regulation of ownership, with the Centers for Medicare and Medicaid services requiring more data and disclosures about nursing home ownership (Centers for Medicare and Medicaid Services, 2023)

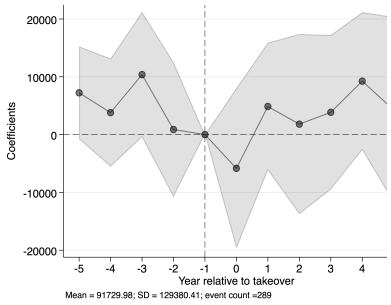
in check. Since Medicare and private pay generally pay more, if a nursing home can keep costs fixed when shifting towards Medicare and private pay residents then their resident profitability will increase. Notably, nursing hours and the prices themselves do not significantly change. The result on unchanging nursing hours seems close to Gupta et al. (2021) whose results suggest no significant change in nursing hours until much later after the takeover. They do not study changes in prices or the payer-type composition. They also find significant increases in facility fixed costs following a takeover, which in our view is likely attributable to the tunnelling of profits which Gandhi and Olenski (2024) find. Our event study also suggests that fixed costs rise following takeovers, but we do not have enough power to conclude statistical significance. In any case, the key point is that we think our hand-coding of takeovers is correct. Our hand-coded takeovers suggest that takeovers of existing facilities, rather than new entry, are the more relevant extensive margin to consider.

Figure A2: Event-study coefficients: Takeover

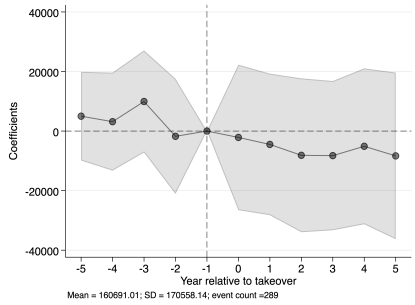
(**a**) Resident Profit



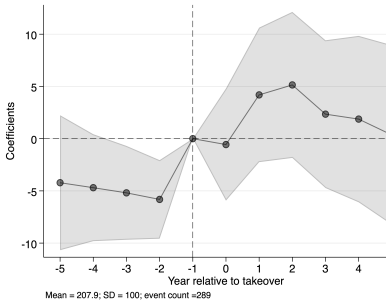
(**b**) Average Total Cost



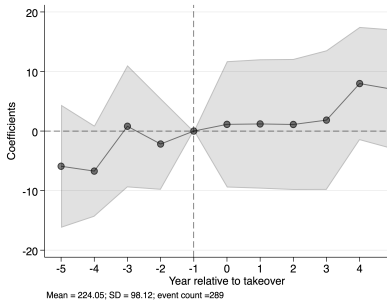
(**c**) Average Variable Cost



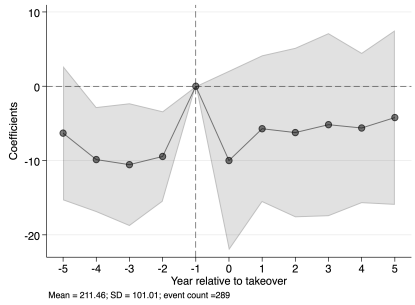
(**d**) Average Medicaid Price



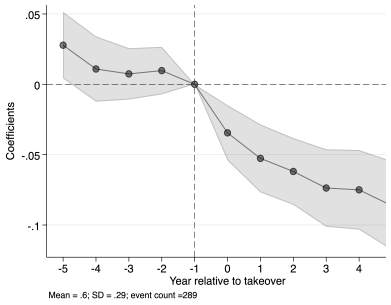
(**e**) Average Medicare Price



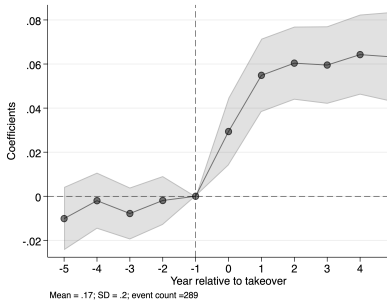
(**f**) Average Private Price



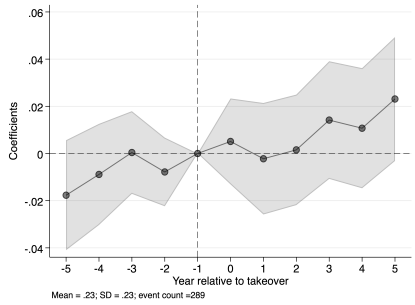
(**g**) Medicaid Share



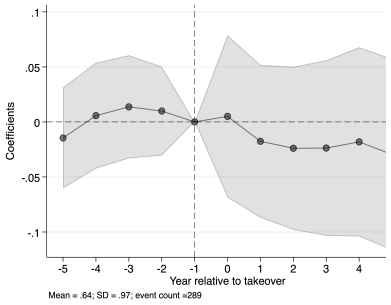
(**h**) Medicare Share



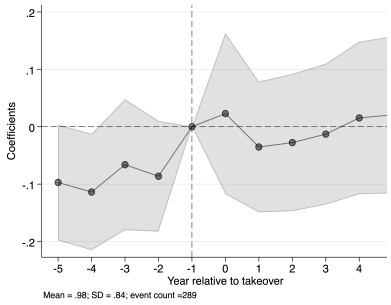
(**i**) Private Share



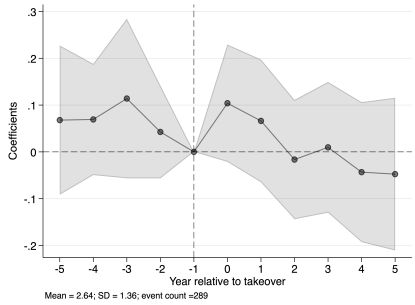
(**j**) Nursing Hours: RN



(**k**) Nursing Hours: LPN



(**l**) Nursing Hours: CNA



Nursing hours are hours per resident per day.

B Supplements to structural estimation

B.1 Full demand parameter estimates

Table B1: Full demand parameter estimates - Part 1

	Baseline	Age (10 yrs)	Los 30-100	Los 100+	Resp. Non-self	Payment Self	Payment Medicaid	Payment Medicare
Distance	-0.1195	-0.2706	-0.1165	-0.1486	-0.1198			
	(0.0009)	(0.0107)	(0.0022)	(0.0041)	(0.0139)			
Distance Sq	-0.0001	0.0500	0.0507	0.1361	0.0833			
	(0.00007)	(0.0048)	(0.0036)	(0.0041)	(0.0080)			
Quality	-1.3256	0.2346	-0.1823	-0.2966	-0.7458		-0.3370	0.6284
	(0.7248)	(0.0025)	(0.0006)	(0.0016)	(0.0081)		(0.0063)	(0.0033)
Price	-0.0248	0.0018	0.0009	-0.0001	-0.0001	-0.0017		
	(0.0051)	(0.00002)	(0.0001)	(0.0001)	(0.0001)	(0.0002)		
Inertia	3.9590							
	(0.0037)							
NFP	-4.0118	0.5142	-0.2035	0.3653	-0.2146		-0.5269	0.8266
	(0.2112)	(0.0007)	(0.0010)	(0.0035)	(0.0011)		(0.0109)	(0.0039)

Table B2: Full demand parameter estimates - Part 2

	DRS	CPS	ADL	CLN	Has Alz.	Female	Non-white	Bach. Educ.
Quality	-0.0166	-0.0669	0.0785	0.2321	-0.3193	0.0265	-0.1729	0.3231
	(0.0025)	(0.0012)	(0.0007)	(0.0004)	(0.0004)	(0.0019)	(0.0038)	(0.0037)
Price								0.0011
								(0.0002)
NFP	0.1052	-0.0115	-0.0876	0.0230	-0.0510	0.1801	-0.4103	0.6574
	(0.0003)	(0.0002)	(0.0002)	(0.0001)	(0.0012)	(0.0001)	(0.0020)	(0.0020)

Note: These results use individual-level data from 2000-2010 in California. Standard errors were calculated by block bootstrap (drawing nursing homes) for 400 iterations.

B.2 1st Stage F-statistics for Supply

Table B3: 1st Stage F-Statistics for Supply

	Quality	Total Quantity	Medicaid Quantity	Medicare Quantity
FP	16.02	73.16	58.07	21.93
NFP	9.95	77.27	87.80	25.72

B.3 Interpreting Supply Estimates Through Cost Report Data

In Table B4, we study how average spending from different cost-centers reported in the cost report data vary across FP and NFP providers. In each column we regress the average cost in each cost center on a NFP indicator with zip and year fixed effects. In the bottom row, we also present the relative sizes of these cost centers by showing their average relative proportions.

The first column focuses on the skilled nursing cost center, which captures spending arising directly from the provision of skilled nursing care.[58] Our estimate of $11.42 suggests that NFP providers do report higher spending on components of skilled nursing care other than the licensed nursing wage bill. Next, ancillary covers a range of health-related expenses that support the provision of skilled nursing care. This includes for example spending on therapists, such as physical therapy and patient supplies. NFP providers appear to spend less on ancillary care compared to FP providers (column 2). It is also noticeable how much ancillary care grows with the proportion of Medicare payers - consistent with the relatively high cost of serving Medicare payers. Finally, non-medical aggregates all non nursing care cost centers such as housekeeping and food. Essentially, this represents costs related to accommodation and living. NFP providers spend around $8 per bed-day more than FP providers in this cost center, so this is likely to be another driver of cost differences between FP and NFP providers. These correlations are suggestive of how FP providers could have a significantly lower cost of serving Medicaid payers compared to NFP providers.

While these data suggest that CNAs and other dimensions of quality might be important we do not model them explicitly. In the case of CNAs, we did not want to introduce further endogenous variables - our supply model is already quite demanding. Pooling them with licensed nurses is also inappropriate as CNAs typically perform tasks which are not medical in nature, while licensed nurses do. For non-medical quality, the issue is that we do not observe the underlying inputs at all. Still, these factors are accounted for in costs indirectly through cost matching. On the demand side, they are captured by the preference for NFP providers, which for some populations is material and significant.

[58] We deduct the wage bill of licensed nurses (registered nurses and licensed practical nurses) as the costs of these are directly captured in the model. The most notable component in this remainder would be the wage costs of certified nursing assistants. These are lower qualified nurses whose functions tend to focus on non-medical aspects of care and represent a substantial proportion of costs. CNAs do not show up in our cost function explicitly but are a part of variable costs in the data through the skilled nursing cost center.

Table B4: Factors driving average cost differences

Cost Center	Skilled nursing minus licensed	Ancillary	Non-Medical
NFP	11.42	-5.44	8.00
	(1.47)	(0.93)	(1.01)
Prop. Mcare	15.21	84.86	-0.33
	(5.13)	(3.81)	(2.71)
Prop. Mcaid	-4.34	-12.41	-6.86
	(2.21)	(1.92)	(1.43)
Av. Proportion of Costs (%)	39.6	18.0	21.5

Note: Regressions include zip and year fixed effects with standard errors clustered at the facility level. These results use facility-level data from 2000-2010 in California.

B.4 Additional Results to Counterfactuals

Table B5: Firm Level Effects from Takeovers

			Mean	25th Pct	Median	75th Pct
Minimal Change	Price	Remaining FP	1.51	-0.65	0.58	2.15
		Takeover NFP	7.39	0.69	7.18	14.90
		Existing NFP	1.46	-0.89	-0.39	2.86
	Quality	Remaining FP	0.011	-0.001	0.001	0.004
		Takeover NFP	-0.028	-0.035	-0.028	-0.018
		Existing NFP	0.014	$\sim$0	0.001	0.005
Half NFP	Price	Remaining FP	11.46	-4.71	1.68	11.43
		Takeover NFP	55.38	14.95	45.74	95.96
		Existing NFP	6.23	-6.19	-1.61	6.51
	Quality	Remaining FP	0.005	-0.019	$\sim$0	0.018
		Takeover NFP	-0.265	-0.334	-0.260	-0.231
		Existing NFP	0.030	$\sim$0	0.005	0.036
Full NFP	Price	Remaining FP	23.30	-8.84	2.37	17.37
		Takeover NFP	101.04	27.54	87.38	162.63
		Existing NFP	5.11	-17.48	-2.31	5.13
	Quality	Remaining FP	0.007	-0.036	0.003	0.025
		Takeover NFP	-0.531	-0.592	-0.544	-0.485
		Existing NFP	0.036	0.001	0.010	0.052

Note: This table summarizes the firm-level impacts of the counterfactuals under different parameterizations of the takeover. Each statistic is taken from the distribution of outcomes across markets where in each market, the lowest quality FP is taken over by a NFP provider. Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. Under the minimal change parameterization, all parameters of the FP provider being taken over are 1/20th a step closer to the NFP parameters and 1/20th of the NFP demand parameters are added. In the Half NFP parameterization, all parameters of the FP provider being taken over are 1/2 between FP versus NFP parameters and 1/2 of the NFP demand parameters are added. In the Full NFP parameterization, all parameters of the FP provider are changed to NFP parameters and full demand parameters are added.

Table B6: Changes in CS excluding large change markets

	Mean	25th Pct	Median	75th Pct
Minimal Change	-0.19	-0.24	-0.13	-0.02
Half NFP	-1.19	-1.89	-0.70	-0.11
Full NFP	-1.81	-3.11	-0.97	-0.22

Note: This table summarizes the impacts of the counterfactuals on changes in CS under different parameterizations of the takeover. Each statistic is taken from the distribution of outcomes across markets where in each market, the lowest quality FP is taken over by a NFP provider. Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. Under the minimal change parameterization, all parameters of the FP provider being taken over are 1/20th a step closer to the NFP parameters and 1/20th of the NFP demand parameters are added. In the Half NFP parameterization, all parameters of the FP provider being taken over are 1/2 between FP versus NFP parameters and 1/2 of the NFP demand parameters are added. In the Full NFP parameterization, all parameters of the FP provider are changed to NFP parameters and full demand parameters are added.

Table B7: Markets with Large Increases from Takeovers - Part 1

			All markets	Large change markets
Minimal Change	Takeover NFP Change	Price	7.38	14.73
		Quality	-0.028	-0.01
	Remaining FP Change	Price	1.51	3.90
		Quality	0.011	0.048
	Existing NFP Change	Price	1.46	7.11
		Quality	0.014	0.088
Half NFP	Takeover NFP Change	Price	55.38	77.01
		Quality	-0.265	-0.245
	Remaining FP Change	Price	11.46	-0.70
		Quality	0.005	0.059
	Existing NFP Change	Price	6.23	4.05
		Quality	0.030	0.093
	Market Features	Baseline Total Bed-days Served ('000)	358.2	837.8
		Takeover Baseline Prop Private Payer	0.239	0.327

Note: This table summarizes the firm-level impacts of the counterfactuals under different parameterizations of the takeover for 'All markets' and 'Large change markets' - where the change in total quantity in the market was larger than 100 bed-days. Each firm-outcome statistic is taken from the distribution of outcomes across markets where in each market, the lowest quality FP is taken over by a NFP provider. Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. Under the minimal change parameterization, all parameters of the FP provider being taken over are 1/20th a step closer to the NFP parameters and 1/20th of the NFP demand parameters are added. In the Half NFP parameterization, all parameters of the FP provider being taken over are 1/2 between FP versus NFP parameters and 1/2 of the NFP demand parameters are added.

Table B8: Markets with Large Increases from Takeovers - Part 2

			All markets	Large change markets
Full NFP	Takeover NFP Change	Price	101.03	211.37
		Quality	-0.531	-0.551
	Remaining FP Change	Price	23.30	-3.88
		Quality	0.007	0.082
	Existing NFP Change	Price	5.11	0.47
		Quality	0.036	0.105
	Market Features	Baseline Total Bed-days Served ('000)	358.2	837.8
		Takeover Baseline Prop Private Payer	0.239	0.327

Note: This table summarizes the firm-level impacts of the counterfactuals under different parameterizations of the takeover for 'All markets' and 'Large change markets' - where the change in total quantity in the market was larger than 100 bed-days. Market Features are reproduced here as in Table B7 for convenience. Each firm-outcome statistic is taken from the distribution of outcomes across markets where in each market, the lowest quality FP is taken over by a NFP provider. Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. In the Full NFP parameterization, all parameters of the FP provider are changed to NFP parameters and full demand parameters are added.